SC^3: A Multi-Solvent Solubility Challenge with Calibrated Aleatoric Ground Truth

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Abstract

We present SC^3, a multi-solvent experimental solubility benchmark built from BIGSOLDB v2.1. Aggregated literature data mixes unit conventions, duplicated publications, and ambiguous stereochemistry; we describe a reproducible pipeline—raw audit, SMILES canonicalization that preserves stereo and distinct tautomers where relevant, and a staged cleaning waterfall—that yields a single unified measurement table suitable for modelling. We estimate an *aleatoric* error floor by merging non-independent sources, fitting thermodynamic curves per independence group, and comparing interpolated values across groups, revisiting the widespread interpretation of ~ 0.6–0.8 log S figures from inter-lab studies. Pairs with multiple independent sources define nested Gold / Silver / Bronze evaluation tiers with consensus targets and per-point uncertainties σ. These pair with standard train/eval splits on high-frequency solvents and an out-of-distribution solvent split; we advocate per-solvent RMSE (PS-RMSE) and uncertainty-aware Z-RMSE alongside aggregate RMSE so that headline numbers are not dominated by a handful of solvents. We benchmark diverse paradigms—physical baselines, boosted trees on descriptors and fingerprints, neural networks, graph networks, and pretrained molecular encoders—under fixed splits and seeds. SC^3 is intended both as a fair comparison suite and as infrastructure for analyses that require calibrated measurement noise rather than noise-free labels.

1 Introduction

Solubility is an intrinsic characteristic of a compound, usually defined as maximal achievable equilibrium concentration of a compound (solute) in a certain solvent under given conditions (temperature, pressure). The knowledge of solubility values for a single solute-solvent pair dovetails with its further applications in drug design [Ma et al. [2025], Bolla et al. [2022]], organic synthesis pipelines [Reichardt and Welton [2011], Diorazio et al. [2016], Tu et al. [2025]] or crystallization studies [Sheikholeslamzadeh et al. [2012], Mendis et al. [2022]]. Namely, water solubility usually guides the preparation of pharmaceutically active substances, whereas organic solubility determines solvent choice in flow and batch synthesis, affecting entire sectors of the economy.

Thus, any *a priori* knowledge on solubility of organic compounds in organic solvents and water will notably accelerate the development of the abovementioned technological processes. For this purpose, many predictive approaches were developed so far to estimate solubilities across diverse subsets of compounds [Fowles et al. [2025]], achieving several noticeable milestones. Boobier *et al.* incorporated a set of computational physicochemical descriptors, which enabled creation of simple yet performative and predictive machine learning models [Boobier et al. [2020]]. Later, Attia *et al.*[Attia et al. [2025]] and Al Ibrahim *et al.*[Al Ibrahim et al. [2025]] developed neural network architectures which performance is approaching the so-called aleatoric limit (inherent error caused by

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inconsistencies in the data). Numerous other approaches incorporating large language models, graph transformers or even boosting algorithms with carefully distilled set of features can reach close to this limit [Jung et al. [2025], Broadbent et al. [2025], Ramani et al. [2026]]. It should be highlighted that most of these methods rely on large-scale experimental datasets, with AqSolDB [Sorkun et al. [2019]], BigSolDB2.0 [Krasnov et al. [2025]] and MixtureSolDB [Malikov et al. [2026]] being largest congeners up to date.

Still, although notable success was achieved in terms of models performance, severe discrepancies in data collection, cleaning and sanitation as well as in validation strategies across different studies hinder direct comparison of methodologies, what hampers fair assessment of the realm of developed approaches. Moreover, imprecise data handling can limit our insights in terms of actual generalization capability of ML models, their intrinsic interpretability and real-life application potential [Llompart et al. [2024]]. To this end, a thorough benchmark study based on a large-scale experimental database is needed; BIGSOLDB, currently the largest literature-aggregated solubility resource, provides that foundation.

In this work we introduce SC^3, an open multi-solvent benchmark derived from BIGSOLDB v2.1 after systematic auditing and cleaning (§2), with an empirically grounded aleatoric floor and tiered consensus evaluation sets carrying per-point uncertainties (§3), leakage-checked train/eval/OOD/tier splits (§4), and a metric suite tailored to multi-solvent regression (§5). We train and evaluate a broad catalogue of baselines under fixed protocols (§6–7), analyse data scaling relative to the calibrated floor (§9), interpret models where labels allow meaningful attribution (§11), and study transfer from an adjacent quantum-chemistry solvation task (§10). Source code, split files, and regeneration scripts accompany the paper.

2 Data curation

SC^3 is derived from BIGSOLDB v2.1 [Krasnov et al., 2025]: 112 465 mole-fraction solubility measurements spanning 1 525 solutes, 218 solvents, 1 687 literature sources, and 243.15–425.77 K. An initial pass revealed 9 DOIs with systematic errors (§2.4), 46 DOIs in which data was repeated from other sources (§2.1), and transcription mistakes between papers (§2.3). Such findings motivated a pipeline consisting of three phases: a *raw audit* phase (§2.1), a *canonicalization* phase (§2.2), and a *cleaning waterfall* phase (§2.4), which reduced the raw table to 101 535 measurements. It is on this dataset that we perform our calculations for the aleatoric limit (§3) and benchmark splits (§4).

2.1 Raw-data audit

Our first phase in the pipeline is to inspect the raw BIGSOLDB dataset (which in addition to the solubility dataset, also provides solvent density data for solvents as a supplementary dataset). Each row carries solute and solvent identifiers (`SMILES_Solute`, `SMILES_Solvent`), temperature, mole fraction x, and a reported $\log_{10} S$ (mol/L) (which is either derived from x and solvent density, or is reporter directly in the literature). Apart from one invalid solute SMILES (which is parsed as a polymer and subsequently ignored, see 2.4), all 1 525 solute SMILES pass `RDKit` (a cheminformatics library for parsing and manipulating SMILES).

Internal consistency of $\log_{10} S$. 3 187 rows (2.8 %) carry `NaN` in the reported $\log_{10} S$ column. For the 109 278 rows with finite mole fraction, temperature, and a non-missing reported $\log_{10} S$, we reconstruct $\log_{10} S$ from the mole fraction x using the standard relation between mole fraction and molar solubility. We fit the provided solvent-density data per solvent to a linear law $\rho(T) = aT + b$ ($\mathrm{g\,cm^{-3}}$), and evaluate ρ at each row's temperature; solvent molar mass M_w is obtained from the solvent SMILES via `RDKit`. Specifically, we evaluate:

$$\log_{10} S = \log_{10}\left(\frac{x}{1-x} \cdot \frac{\rho(T)\,\cdot\,10^3}{M_w}\right),$$

and we compare the result to the reported $\log_{10} S$. Residuals are negligible (median $|\mathrm{residual}| = 9.9 \times 10^{-6}$, $P_{99} = 1.25 \times 10^{-3}$); only 8 rows exceed 0.01 log S, all from a single DOI (`10.1016/j.molliq.2017.02.075`, β-alanine in methanol), which we add to the bad-DOI list (§2.4). This agreement is what licenses us to use the *same* closed form in the cleaning waterfall to compute $\log_{10} S$ for the rows in which $\log_{10} S$ is `NaN` (§2.4, step W5), wherein only 14 rows are dropped because no density route returns a positive ρ for that solvent at that temperature.

Cross-DOI bit-exact matches. We group rows by (canonical solute, canonical solvent, T rounded to 0.1 K, $\log_{10} S$ rounded to four decimals): there are 86 such four-tuples reported by two or more DOIs at identical values, which affects 172 rows across 46 DOIs — this is an artifact of data duplication between sources, and is consumed downstream when we build independence groups for the aleatoric analysis (§3.1).

2.2 Canonicalization

We canonicalize all solute and solvent SMILES with `RDKit`, preserving stereochemistry (chirality and E/Z geometry) and we explicitly choose to *not* enumerate tautomers; this leaves all 1 525 raw solute SMILES distinct. Alternative policies that strip stereochemistry or enumerate tautomers were considered and rejected, because the merges they induce conflate compounds whose measured solubilities disagree by 5–20× the aleatoric floor. The candidate policies, an empirical audit of those merges, and the chemical interpretation are deferred to Appendix A.

2.3 Manual corrections

The source-integrity analysis (§3.1) flagged 22 DOIs with mean inter-lab deviation $\geq$ 0.6 log S against peer consensus. Manual review of those papers identifies four transcription bugs that we correct in place (Table 1), recovering 58 rows that the bad-DOI filter (W1) would otherwise drop wholesale.

Table 1: Manual corrections applied before the cleaning waterfall 2.4.

Class	Correction	DOI	rows
C1	paracetamol / water: four x values replaced with audited values	`10.1016/j.molliq.2020.113867`	4
C2	$\log_{10} S$ shifted by $+1$ ($\times 10$ unit correction)	`10.1016/j.fluid.2013.09.018`	20
C2	$\log_{10} S$ shifted by $+1$ ($\times 10$ unit correction)	`10.1016/j.molliq.2013.06.011`	10
C3	ethanol $\leftrightarrow$ ethyl acetate labels swapped, $\log_{10} S$ re-derived	`10.1016/j.fluid.2015.07.038`	24

2.4 Cleaning waterfall

After canonicalization and manual corrections applied to the raw data, the remaining cleaning is a seven-stage waterfall, detailed below (Table 2).

Table 2: Cleaning waterfall. Each stage is applied in order to the manually corrected and canonicalized data.

Step	Filter	Rows after	Removed
	Input (canonicalized, manually corrected raw)	112 465	—
W1	remove 9 bad DOIs (see text)	112 019	446
W2	invalid / polymer solvent SMILES (`"-"`)	111 705	314
W3	salts / mixtures (`.` in solute SMILES)	101 663	10 042
W4	$M_w > 1000$ Da	101 634	29
W5	$\log_{10} S$ recovered for 2 617 `NaN` rows; 14 unrecoverable	101 620	14
W6	$\log_{10} S \notin [-15, 2]$	101 575	45
W7	intra-DOI near-duplicate dedupe (T rounded to 0.1 K)	**101 535**	40

W1 — bad DOIs. The nine excluded DOIs come from three sources: five from the manual transcription-bug review of §2.3, three from the per-DOI consensus-deviation audit of §3.1, and the single $\log_{10} S$-residual DOI identified in §2.1 (Table 3).

W3 — salts and mixtures. The 10 042 dropped rows correspond to 193 distinct multi-component solute SMILES. A single $\log_{10} S$ label is not directly comparable across salt and neutral entries because their dissolution thermodynamics differ structurally; we exclude them here and leave a salt-aware split for future work **[TODO: FIND CITATION]**.

Table 3: Nine DOIs removed at W1.

Source of flag	DOI

Manual review (§2.3)	`10.1016/j.fluid.2011.09.033`
	`10.1021/jced.9b00728`
	`10.1021/jced.4c00179`
	`10.1021/jced.6b00009`
	`10.1016/j.molliq.2022.119759`

Consensus-deviation audit (§3.1)	`10.1021/je4000718`
	`10.1016/s1004-9541(08)60201-3`
	`10.1016/j.molliq.2016.11.036`

$\log_{10} S$-residual outlier (§2.1)	`10.1016/j.molliq.2017.02.075`

W5 — $\log_{10} S$ recovery. We re-use the same closed form as in §2.1, resolving ρ at the row's $(\text{solvent}, T)$ from one of three sources: (i) the experimental value from BIGSOLDB v2.1's solvent-density table when present; (ii) else the (a, b) linear-fit coefficients from §2.1; (iii) else the `thermo` library (a thermophysical property library for estimating density from temperature and pressure). Of the 2 631 rows that reach step W5 with `NaN` $\log_{10} S$ (§2.4, step W5), 2 617 recover a finite $\log_{10} S$; the remaining 14 cannot be assigned a positive ρ by any tier and are dropped.

W7 — intra-DOI dedupe. All 40 removed rows come from a single DOI (`10.1016/j.jct.2020.106137`) that reports each nominal temperature twice ($T = x.15$ and $x.16$ K) with identical $\log_{10} S$ values, which we attribute to a rounding artifact.

Final cleaned dataset. After canonicalization (Option D), the four manual corrections, and the seven-stage waterfall above, SC^3 (Table 4) contains 101 535 measurements covering 1 327 canonical solutes, 206 solvents, and 1 493 DOIs, with 746 (solute, solvent) pairs measured by two or more independent DOIs (the multi-source pool that the aleatoric analysis of §3 consumes).

Table 4: Headline figures of the cleaned SC^3 dataset.

measurements	**101 535**
unique solutes (canonical)	**1 327**
unique solvents	**206**
unique DOIs	**1 493**
temperature range	243.15–425.77 K
$\log_{10} S$ range	-7.56–$+1.99$
(solute, solvent) pairs with ≥ 2 DOIs	**746** (6.8 %)

3 Aleatoric limit

With a cleaned dataset in hand (§2), the next question is what precision its labels allow. Solubility is an experimental quantity, measured in different laboratories with different instruments and techniques, so every label carries measurement noise; the disagreement between independent measurements of the same (solute, solvent, T) point is a lower bound on model error that no predictive method can cross. This *aleatoric limit* is the benchmark's ceiling.

The commonly cited 0.6–0.8 log S figure from Palmer and Mitchell [2014] has been used to argue that solubility prediction is "solved" once models reach this range. We show that (i) this number is the 90–95$^{\text{th}}$ percentile of a heavy-tailed inter-lab distribution rather than the expected error, and (ii) in expectation inter-lab disagreement is about $6\times$ smaller.

We build the aleatoric floor in three stages. §3.1 assembles *independence groups* of DOIs by merging detectable copies; §3.2 fits a temperature-dependent model to each group for every (solute, solvent) triple it reports; §3.3 defines and computes the aleatoric limit $\varepsilon_{\text{aleatoric}}$ from Apelblat-interpolated comparisons across those independence groups. Extended threshold sensitivity and Apelblat diagnostic figures are in Appendix B.

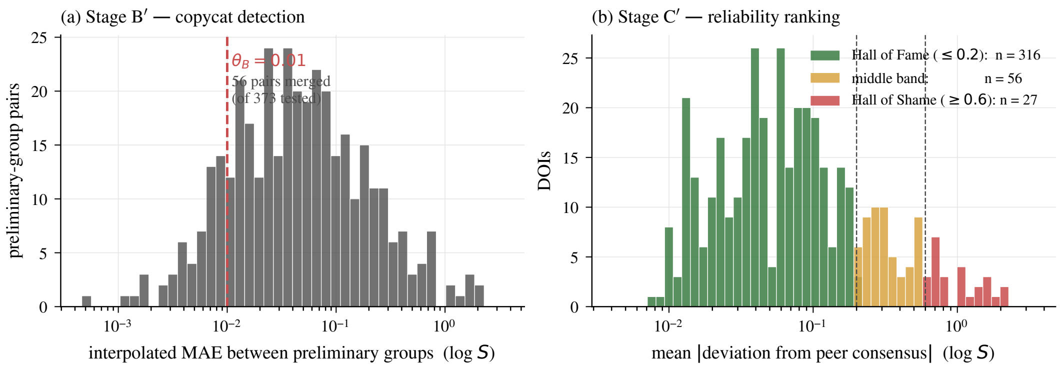

Figure 1: **(a) Stage B′ copycat detection.** Distribution of interpolated pair-MAE between preliminary-group pairs sharing at least one (solute, solvent) pair. Dashed line = $\theta_{B'} = 0.01$ log S cut. **(b) Stage C′ reliability ranking.** Distribution of per-group mean absolute deviation from peer consensus on Apelblat-interpolated curves. Hall of Fame (≤ 0.2), middle band, and Hall of Shame (≥ 0.6) are coloured; dashed vertical lines are the cut-offs.

3.1 Source integrity

Because a single research group sometimes re-publishes the same solubility data across multiple papers, treating every DOI as an independent measurement inflates the apparent sample size and biases inter-lab statistics toward zero. We aggregate DOIs into *independence groups* using two stages of copycat detection, followed by a reliability ranking against the consensus of other groups.

Stage A — bit-exact. DOIs that report an identical four-tuple $(\mathrm{solute}, \mathrm{solvent}, \mathrm{round}(T, 1), \mathrm{round}(\log_{10} S, 4))$ are unioned deterministically. No threshold is required: bit-exact agreement to the fourth decimal across independent DOIs is functionally impossible as a measurement outcome. In the cleaned dataset, 86 such tuples occur, affecting 172 rows across 46 distinct DOIs; this yields 21 inter-DOI unions.

Stage B' — interpolated gray zone. Many copycats publish a re-scaled or slightly relabelled version of the same data and do *not* satisfy bit-exact equality, so Stage A misses them. After fitting per-group Apelblat / van't-Hoff curves (§3.2), for every preliminary-group pair sharing a $(\mathrm{solute}, \mathrm{solvent})$ pair we compute the MAE between their fitted curves evaluated on a uniform 1 K reference grid in the intersection of their fit ranges (overlap ≥ 5 K required). Pairs whose weighted MAE across all shared (solute, solvent) pairs falls below $\theta_{B'} = 0.01$ log S are unioned. This yields 55 additional unions, producing 1 415 final independence groups (down from 1 493 DOIs): a size histogram of 1 348 singletons, 57 pairs, 8 triples, 2 quadruples.

Figure 1(a) shows the distribution of these interpolated pair-MAEs. The $\theta_{B'} = 0.01$ cut sits in a sparse region of the distribution: only 55 of 373 tested preliminary-group pairs fall below it, while the bulk of the mass lies well above at 0.02–1 log S (genuine inter-lab disagreement).

We adopt $\theta_{B'} = 0.01$ log S as the copycat-merge threshold: below this value additional unions are scarce and $\varepsilon_{\mathrm{aleatoric}}$ moves slowly, while just above it the multi-source pool collapses (Figure 28, Appendix B.1).

Stage C' — interpolated reliability. For each final group, we evaluate its fitted curves at a uniform 1 K grid on the intersection of all groups' fit ranges for every (solute, solvent) pair where the group co-appears. At each grid point the *consensus* is the mean of the *other* groups' fitted values; the group's local deviation is the absolute difference between its own value and the consensus. Averaged across all grid points the group participates in, this produces a single reliability score per group. Groups scoring ≤ 0.2 log S form the *Hall of Fame*; groups scoring ≥ 0.6 join the *Hall of Shame* (Figure 1(b)). Of 1 415 final groups, 399 have enough cross-group overlap to be tested; 316 land in the Hall of Fame and 27 in the Hall of Shame.

3.2 Thermodynamic curve fits

To compare laboratories that do not share exact measurement temperatures, we fit a smooth thermodynamic model to every $(\text{solute}, \text{solvent}, \text{group})$ triple with ≥ 3 temperature points:

$$\log_{10} S \;=\; A \;+\; \frac{B}{T} \;+\; C\,\ln T \qquad \text{(Apelblat)}, \tag{1}$$

and a reduced form to triples with exactly two temperature points:

$$\log_{10} S \;=\; A \;+\; \frac{B}{T} \qquad \text{(van't Hoff)}. \tag{2}$$

Triples with a single temperature point (507, $\sim$ 4.5 %) cannot be fitted and are excluded from aleatoric interpolation. Fits with $R^2 < 0.80$ or RMSE > 0.30 log S (16) are flagged as low-quality and also excluded. The remaining 11 063 fits cover $\geq$ 95 % of the cleaned data. Fit-quality and temperature-coverage distributions are shown in Appendix B.2 (Figure 29).

3.3 Aleatoric limit

With independence groups and per-group curves established, the aleatoric limit is a single pair-level statistic on this structure.

Mathematical framework. For an $(\text{solute}, \text{solvent})$ pair at temperature T, the true solubility $\log_{10} S^*(T)$ is unknown. Independence group i observes $y_i(T) = \log_{10} S^*(T) + \varepsilon_i(T)$, where ε_i is measurement noise plus any systematic lab effect. Given fitted curves $f_i(T)$ and $f_j(T)$ for groups i and j on a reference grid $\mathcal{T}_{ij} = [T_{ij}^{\min}, \ldots, T_{ij}^{\max}]$ at $\Delta T = 1$ K, the interpolated inter-group disagreement at cell T is

$$\varepsilon_{ij}(T) \;=\; \big|\, f_i(T) - f_j(T) \,\big|, \tag{3}$$

and the *group-pair* MAE is $\mathrm{MAE}_{ij} = \mathrm{mean}_{T \in \mathcal{T}_{ij}}\, \varepsilon_{ij}(T)$. Per (solute, solvent) pair p with independence groups $\mathcal{G}_p$, the *pair* MAE averages over all within-pair group pairs:

$$\mathrm{MAE}_p \;=\; \frac{1}{\binom{|\mathcal{G}_p|}{2}} \sum_{i<j \in \mathcal{G}_p} \mathrm{MAE}_{ij}. \tag{4}$$

The aleatoric limit is the *mean* pair MAE across the multi-source pool $\mathcal{P}$:

$$\boxed{\varepsilon_{\text{aleatoric}} \;=\; \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathrm{MAE}_p.} \tag{5}$$

Two aspects warrant emphasis. First, $\varepsilon_{\text{aleatoric}}$ is an *error in expectation* over pairs and group-pair comparisons, not a summary of the heavy tail (reported separately below). Second, we evaluate *every* cell via the fitted curve (never a measured value) so the comparison is uniform in character and not biased toward pairs that happen to share exact measured temperatures.

Primary vs inclusive. The Hall-of-Shame groups identified in §3.1 consist overwhelmingly of systematic publication errors (unit swaps, density errors, silent transcription mistakes) rather than random measurement noise. They are excluded from tier consensus labels (§4.1). For the aleatoric limit, we report both variants: a *primary* value that excludes HoS groups (the floor relevant to benchmark performance, because tier labels use the same pool), and an *inclusive* value that retains them (useful when comparing against the broader literature).

The 95 % CI on $\varepsilon_{\text{aleatoric}}$ (primary) excludes zero by about 7 standard errors — the floor is precisely characterised.

Table 5: Primary and inclusive $\varepsilon_{\text{aleatoric}}$. 95 % confidence intervals are from 5 000 bootstrap resamples over the multi-source pair pool.

	n pairs	mean = $\varepsilon_{\text{aleatoric}}$	95 % CI	median	P_{90}	P_{95}	RMSE
Primary (HoS-excluded)	481	**0.106**	[0.093, 0.120]	0.046	0.258	0.385	0.182
Inclusive	511	0.158	[0.132, 0.186]	0.050	0.385	0.627	0.356

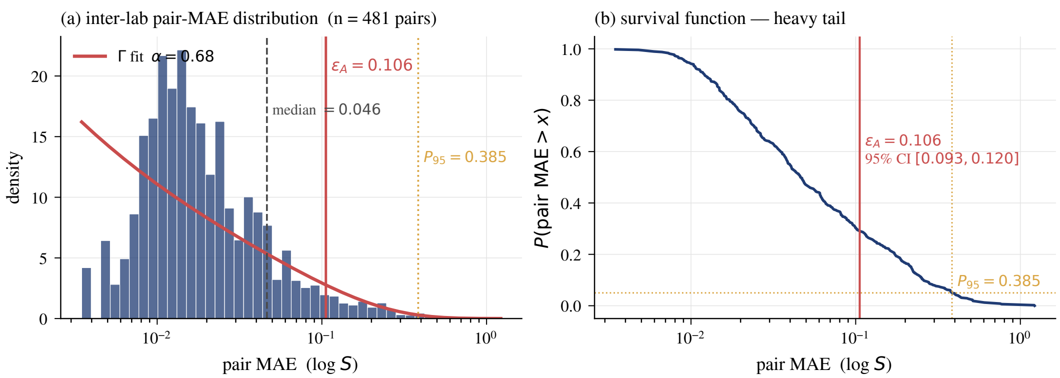

Figure 2: Primary aleatoric distribution (HoS-excluded). **(a)** Density of per-pair MAE on log-x axis with a gamma fit overlay ($\alpha = 0.68$). **(b)** Survival function $P(\text{MAE} > x)$. Heavy right tail: $P_{95} = 0.39$ log S.

Distribution shape. Figure 2 shows the per-pair MAE distribution together with a gamma fit to the atom-level $|\Delta|$ distribution. The gamma shape parameter $\alpha = 0.68$ confirms a sub-exponential (heavy) right tail, which is why the mean (0.106) lies well above the median (0.046). Panel (b) plots the survival function: 5 % of pairs disagree by more than 0.39 log S, and the 99$^{\text{th}}$-percentile pair disagrees by more than 1 log S.

Per-solvent heterogeneity. $\varepsilon_{\text{aleatoric}}$ is not uniform across solvents (Figure 3). DMF has the tightest inter-lab agreement at 0.029 log S across 10 pairs. Water — often assumed to be well-characterised because of the large volume of aqueous data — has the *highest* mean among common solvents (0.110 log S across 65 pairs), driven by a thicker tail rather than a higher typical disagreement (its median is only 0.061). This per-solvent heterogeneity motivates the per-solvent evaluation metric proposed in §5.2 (PS-RMSE): a single global $\varepsilon_{\text{aleatoric}}$ under-characterises measurement quality in difficult solvents and over-characterises it in well-behaved ones.

Reconciliation with Palmer & Mitchell (2014). The 0.6–0.8 log S figure cited in the aqueous-solubility literature is not wrong — it is simply a different summary statistic on the same underlying distribution. Figure 4 overlays our full inclusive atom-level $|\Delta|$ distribution with the Palmer band (shaded). Their number coincides with *our* P_{90}–P_{95} at 0.39–0.63 log S and with our RMSE (0.36), but not with the mean (0.16) or median (0.05). A practitioner who quotes the *expected* inter-lab error — the number models compete against in expectation — should quote $\varepsilon_{\text{aleatoric}} \approx 0.11$ log S; a practitioner who quotes the 95$^{\text{th}}$-percentile worst case — the noise a model faces on its hardest test point — can legitimately still quote 0.6.

4 Benchmark design

With a cleaned dataset (§2) and a calibrated noise floor (§3), we can now build the benchmark itself. SC^3 is designed to answer three distinct generalisation questions simultaneously: can a model predict (i) a new $(\text{solute}, \text{solvent})$ pair in a familiar solvent, (ii) the same solute in a solvent it has never seen, and (iii) a new solute against consensus-calibrated labels with per-point uncertainty σ? Each question corresponds to a separate evaluation set. The construction is: a *tier-test pool* with consensus labels from multi-source agreement, and a *training pool* drawn from the complement that is further split by solvent-frequency into in-distribution (Train, Eval) and out-of-distribution (OOD) sets.

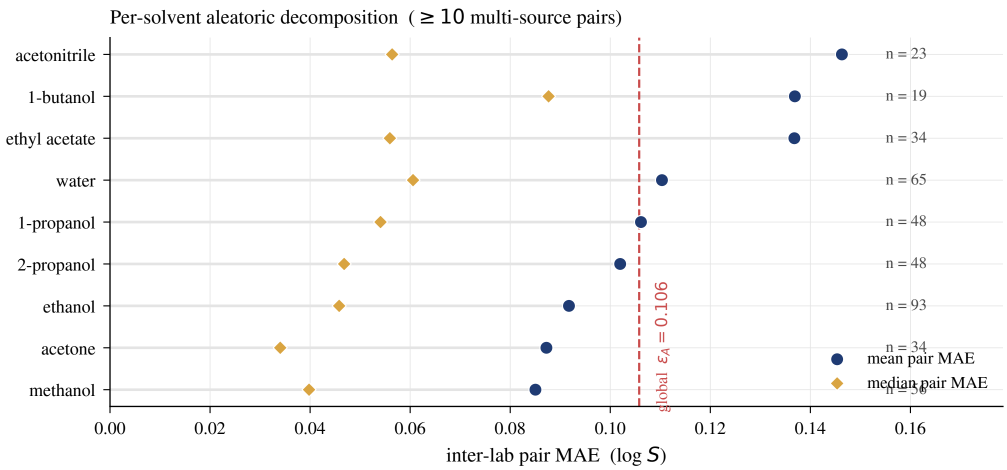

Figure 3: Per-solvent aleatoric decomposition (primary, HoS-excluded, solvents with ≥ 10 multi-source pairs). Lollipops are mean pair MAE; diamonds are median; n is the multi-source pair count. Dashed vertical line is the global $\varepsilon_{\text{aleatoric}} = 0.106$.

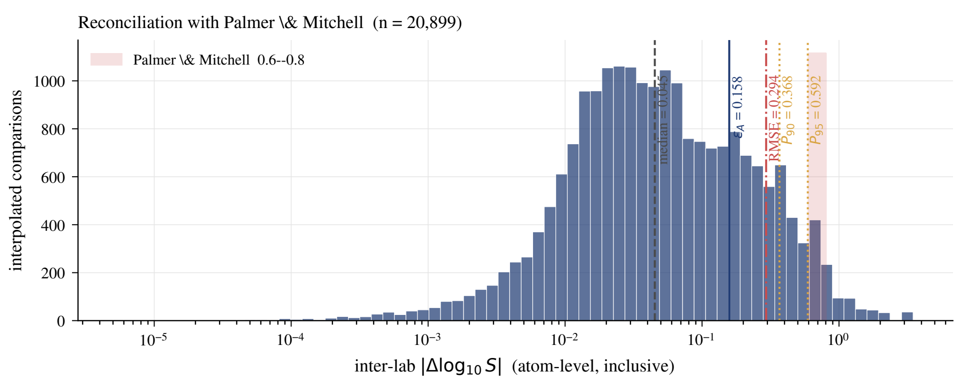

Figure 4: Reconciliation with Palmer and Mitchell [2014]. The inclusive atom-level $|\Delta \log_{10} S|$ distribution ($n = 20\,899$ interpolated comparisons) is plotted on a log-x axis; vertical lines are our summary statistics. The shaded band is Palmer & Mitchell's quoted 0.6–0.8 log S aqueous range, which coincides with our P_{90}–P_{95} and RMSE, not with the expected error.

4.1 Tier construction

The tiers are the benchmark's evaluation with calibrated ground truth and σ. They are built from the 481 multi-source pairs that survive (i) the copycat merging of §3.1, (ii) the Hall-of-Shame exclusion, and (iii) the fit-quality gate of §3.2. For each eligible pair we compute the pair MAE (equation 4) and assign it to every tier whose MAE threshold it satisfies:

Tier	Pair MAE threshold	Meaning
Gold	≤ 0.1 log S	consensus labels disagree by at most $\sim \varepsilon_{\text{aleatoric}}$
Silver	≤ 0.2 log S	within a factor of $\sim 2\varepsilon_{\text{aleatoric}}$
Bronze	≤ 0.5 log S	within a factor of $\sim 5\varepsilon_{\text{aleatoric}}$

The tiers are *nested*: Gold $\subset$ Silver $\subset$ Bronze. The naming is designed so that a **tighter tier label corresponds to a better-characterised ground truth**, not to a harder modelling task. A model evaluated on Gold is being compared against the tightest available reference labels; Bronze is a broader net that keeps more pairs at the cost of looser consensus.

Consensus labels. For each $(\text{solute}, \text{solvent}, T)$ triple in a tier pair, we evaluate every contributing independence group's Apelblat / van't-Hoff fit at T and define the consensus label as the mean of

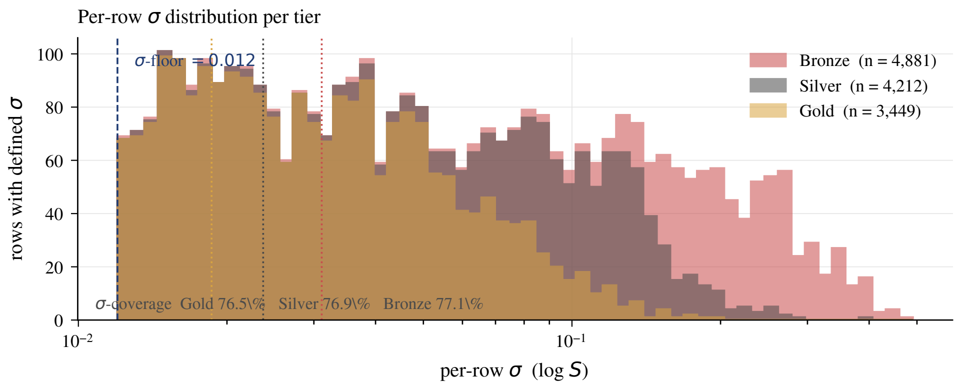

Figure 5: Per-row σ distribution in the three nested tiers, log-σ axis. Dashed vertical line at the σ-floor $= 0.012$ log S; dotted lines mark per-tier medians. Gold's distribution is concentrated between the floor and ~ 0.03 log S; Bronze has a much longer tail out to $\sigma > 0.2$ log S. σ-coverage is near-identical across tiers ($\sim 77\,\%$), so the tail difference is about *quality* of uncertainty, not quantity.

those evaluations, excluding Hall-of-Shame groups. The per-point uncertainty σ is the standard deviation across the same set of evaluations. When fewer than two groups contribute at a given T, σ is undefined and the row carries a hard label with no uncertainty; this happens on 14.4 % of tier rows overall.

Uncertainty floor. A standard deviation from two nearly-identical fits can underestimate true measurement spread. We therefore floor σ at 0.012 log S ($\approx \varepsilon_{\text{aleatoric}}/10$) — the median Apelblat fit RMSE (§3.2) — to prevent a handful of suspiciously-tight σs from dominating Z-normalised losses.

Tier statistics. Table 6 shows size and σ distribution per tier. Median σ ranges from 0.019 (Gold) through 0.024 (Silver) to 0.031 (Bronze); the Gold tail is much shorter ($P_{95} = 0.078$ vs Bronze $P_{95} = 0.245$), reflecting the tighter pair-MAE cap. Figure 5 shows the full σ distribution per tier: the Gold distribution is entirely below Bronze's tail.

Table 6: SC^3 tier composition and σ statistics. Nested: Gold $\subset$ Silver $\subset$ Bronze. σ-coverage is the fraction of tier rows with a defined per-point uncertainty; remaining rows have a hard label only.

Tier	rows	pairs	solutes	solvents	σ-cov.	median σ	mean σ	P_{95} σ
Gold	4 507	335	129	26	76.5%	0.019	0.028	0.078
Silver	5 475	400	141	27	76.9%	0.024	0.042	0.130
Bronze	6 331	469	148	30	77.1%	0.031	0.066	0.245

4.2 Training, evaluation, and OOD splits

Tiers consume only a thin slice of the cleaned pool; the complement forms the training and in-distribution test sets, and the long-tail solvents become the out-of-distribution test set. The 148 solutes that appear in any tier are fully excluded from the training pool at the $(\text{solute}, \text{solvent}, T)$ level — no row containing such a solute survives into Train, Eval, or OOD. The remaining 80 312 rows are split by solvent frequency:

- **Top-25 solvents** (ranked by training-pool row count, covering 85.1 % of the training pool) form the *in-distribution* region. Within this region, 10 % of each solvent's solute list is held out as **Eval**; the remainder is **Train**.
- **Remaining 181 solvents** form the **OOD** set. By construction a model will never have been trained on any of these solvents, which tests whether it can generalise across the solvent axis.

Table 7 summarises the final composition.

Table 7: SC^3 splits and tiers. All six evaluation subsets have zero solute overlap with training (see Table 8). Solvent overlap with Train is intentional for Eval (in-distribution) and tiers (calibrated labels); OOD has zero solvent overlap with Train by construction.

	Role	rows	solutes	solvents	(solute, solvent) pairs
Training pool (80 312 rows after tier exclusion)					
Train	training set	61 403	1 144	25	6 840
Eval	new-pair, in-distribution	6 969	534	25	771
OOD	new-solvent (151 unseen)	11 940	586	161	1 450
Tier test pool (solutes fully absent from training pool)					
Gold	tight consensus, σ-cal.	4 507	129	26	335
Silver	looser consensus, σ-cal.	5 475	141	27	400
Bronze	broadest, σ-cal.	6 331	148	30	469

Generalisation axes exposed. The three evaluation sets test distinct notions of generalisation:

- **Eval** tests *new (solute, solvent) pairs* in already-seen solvents. A model must interpolate across the solute axis.
- **OOD** tests *new solvents* (the 161 long-tail solvents). Solute may have been seen in a different solvent, but the target solvent is unseen. Some solvents have few measurements; we therefore report OOD metrics with the per-solvent split-out explicitly (§5).
- **Gold / Silver / Bronze** test *new solutes* against calibrated consensus labels with σ. Any of the 148 tier solutes has been removed from Train/Eval/OOD, so all tier tests are solute-level out-of-distribution.

Anti-leakage verification. Every overlap that must be zero, is zero (Table 8). Solute overlap between Train and Eval is expected to be positive (520 solutes) because Eval is a held-out slice of the in-distribution solute–solvent grid; pair-level overlap between Train and Eval remains zero by construction of the hold-out.

Table 8: Anti-leakage verification. Every cell reports the count of (solute, solvent) pair overlaps between two splits. Zero everywhere except the two off-diagonal entries that are expected to share solutes but never pairs (shaded). All checks pass.

	Train	Eval	OOD	Gold	Silver	Bronze
Train	—	0	0	0	0	0
Eval	0	—	0	0	0	0
OOD	0	0	—	0	0	0
Gold	0	0	0	—	nested	nested
Silver	0	0	0	nested	—	nested
Bronze	0	0	0	nested	nested	—

FIGURE PLACEHOLDER

`paper/figures/fig_splits_diagram.pdf`

Two-axis schematic of the leakage-hard splits. Horizontal axis = solvent set (25 ID solvents on the left, 161 OOD solvents on the right); vertical axis = solute set (training-pool solutes vs. tier solutes, fully disjoint). Train (top-left), Eval (top-left held-out pairs), OOD (top-right, ID solutes in unseen solvents), and tiers (bottom-left, tier solutes in ID solvents with calibrated σ). Cell row counts: Train 61 403*, Eval* 6969*, OOD* 11 940*, Bronze* 6331*. Use coloured rectangles with row counts inside and a legend identifying each split.*

Figure 6: Three generalisation axes the splits expose. *Eval* tests new (solute, solvent) pairs in familiar solvents; *OOD* tests unseen solvents; the tiers test new solutes in familiar solvents *with calibrated σ*.

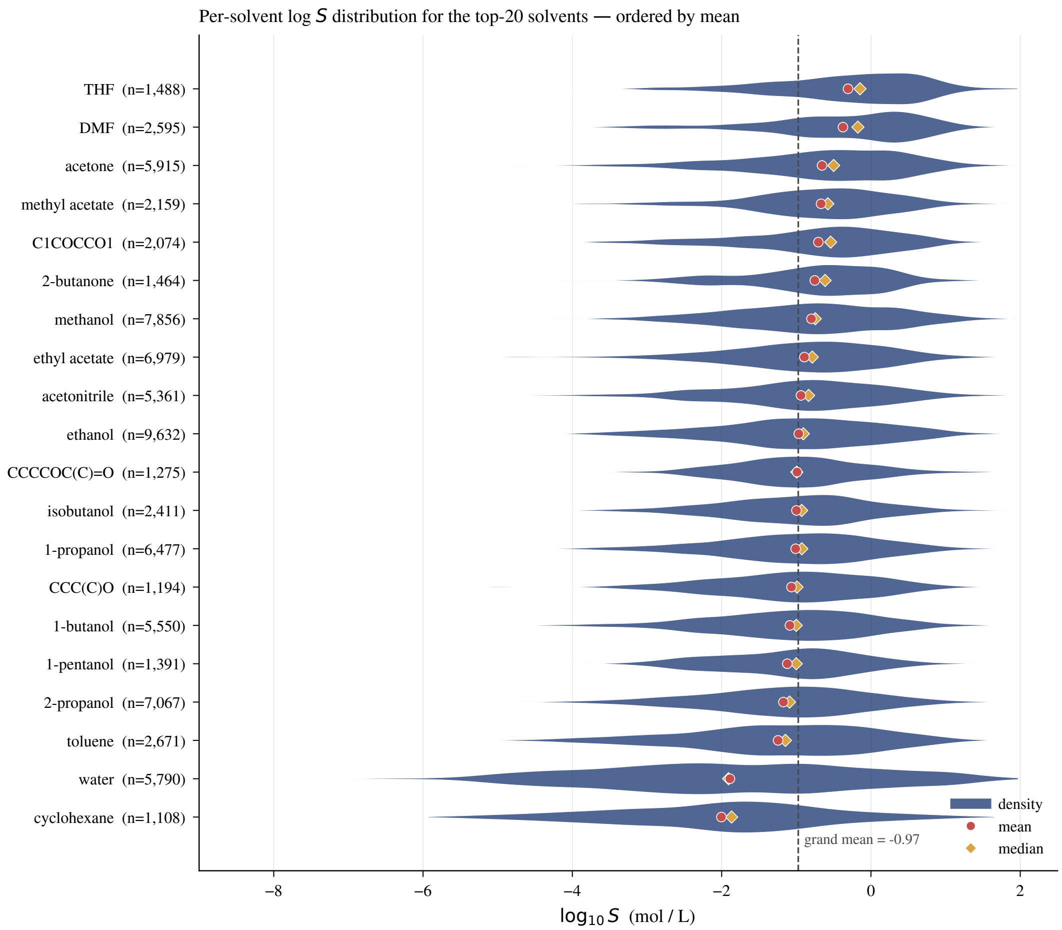

Figure 7: Per-solvent log S distribution (violin) for the top-20 solvents by row count, ordered by per-solvent mean. Red marker = mean, gold diamond = median, dashed line = grand mean. The top-to- bottom mean span is ~ 7 log S; *any* predictor that correctly identifies the solvent absorbs most of that variance for free.

5 Metrics

With the splits in place (§4), we turn to scoring. Reporting aggregate RMSE works for single-solvent regression but fails here for four distinct reasons rooted in the multi-solvent structure of the data. We characterise those reasons empirically on SC^3, then define the five-metric suite we recommend reporting for every model.

5.1 Why multi-solvent is different

Per-solvent location shift. Across the 206 cleaned solvents, per-solvent log S means span **7.98 log units** (minimum -6.93, maximum +1.05) — nine orders of magnitude in equivalent concentration. Figure 7 shows the per-solvent log S distribution for the top-20 solvents by row count, ordered by their per-solvent mean. The densities overlap substantially in the middle of the range (both water and alcohols contain solutes spanning most of the dataset's log S range) but their locations differ by up to seven log units.

Variance decomposition. A one-way ANOVA of cleaned log S on solvent identity attributes **11.7 %** of total variance to the between-solvent location shift ($F = 69.9$ across 206 solvents, $p < 10^{-300}$). On OOD, which is built from the long-tail solvents, the between-solvent fraction rises to **22.1 %** because the long-tail solvents include extremes (e.g., 1,1-dichloroethane with mean log S $= -6.93$). Within each tier the fraction is closer to 9.3–9.9 %.

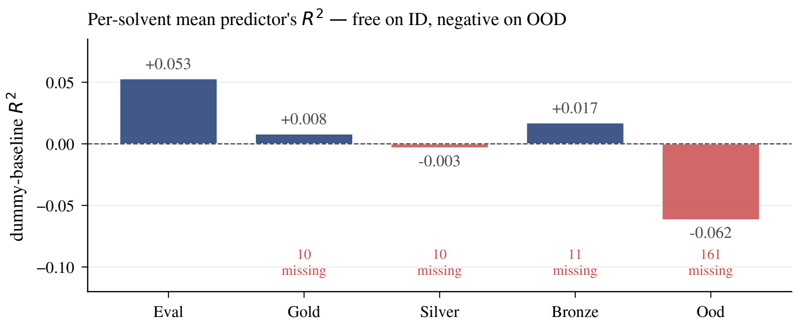

Figure 8: Per-solvent-mean predictor's R^2 across the five evaluation splits. Positive on in-distribution Eval, near-zero on tiers, and *negative* on OOD (161 solvents unseen during training, so the predictor falls back to the grand mean). Motivates a metric suite in which aggregate R^2 or RMSE does not reward solvent identification.

Dummy-baseline $\mathbf{R^2}$. We quantify how much of this variance is "free" for a trivial model. A predictor that returns the *per-solvent training mean* (no solute information, no temperature) achieves the R^2 values in Figure 8. On the in-distribution Eval split, the baseline scores $R^2 = +0.053$ — positive, non-trivial, and achievable by a model that only memorises the 25 training-solvent means. On the three tiers it scores $R^2 \in [-0.003, +0.017]$, near zero but within the ID solvent space. On OOD, where 161 of 161 test solvents are never seen during training and the baseline necessarily falls back to the grand training mean, it **goes negative** ($R^2 = -0.062$) — the grand mean is actively worse than no prediction at all on the long-tail solvents.

Count domination. The top-5 solvents (*ethanol, methanol, 2-propanol, ethyl acetate, 1-propanol*) account for **37.5 %** of the cleaned rows and **44.2 %** of Train; the top-25 account for 84.5 % and cover 100 % of Train / Eval by construction. An aggregate-row metric such as RMSE is therefore, functionally, a metric on those five solvents. Reporting only aggregates would hide per-solvent regression on the long-tail set.

MAPE is broken here. 5.6 % of cleaned rows have $|\log_{10} S| < 0.1$, and 28.5 % have $|\log_{10} S| < 0.5$ — the MAPE denominator is small or zero for a material fraction of the data. MAPE is therefore unbounded for any model that makes ordinary-magnitude absolute errors on these rows and should not be reported as a headline statistic.

Heavy-tailed labels. The $|y - \bar{y}|$ distribution has mean/median ratio 1.19 ($P_{95} = 2.27$, $P_{99} = 3.43$). RMSE is tail-sensitive in a regime this heavy, motivating a median absolute error as a robust complement.

5.2 The SC^3 metric suite

Combining the above, we define five primary metrics, with **PS-RMSE** as the single headline. Definitions are in Table 9.

PS-RMSE. Per-solvent RMSE averaged equally across solvents. This resolves both the variance-decomposition and count-domination problems from §5.1 in one stroke: between-solvent location shifts cancel out of the per-solvent RMSE, and each solvent contributes equally regardless of its row count. For $|\mathcal{S}|$ large enough, PS-RMSE approaches the "within-solvent" RMSE that the ANOVA identifies as the honest prediction-error variance.

Z-RMSE. Error in units of the aleatoric floor. On rows where σ_i is defined (median 77 % of tier rows, Table 6), Z-RMSE $= 1$ means a model is operating at the measurement-noise limit; Z-RMSE $= 10$ means errors are an order of magnitude larger than irreducible label uncertainty. Because SC^3 calibrates σ per point, Z-RMSE is a *per-measurement* comparison to the floor, not a comparison to a single global $\varepsilon_{\text{aleatoric}}$.

Table 9: SC^3 metric suite. n = rows in the split; $|\mathcal{S}|$ = solvents in the split; σ_i = per-row calibrated uncertainty (tiers only). The "applicability" column lists the splits on which each metric is computed. MAPE is reported for diagnostic purposes only.

Metric	Formula	Applicability	Role
RMSE	$\sqrt{\frac{1}{n}\sum_i(\hat{y}_i - y_i)^2}$	all	count-weighted; baseline
MAE	$\frac{1}{n}\sum_i \lvert\hat{y}_i - y_i\rvert$	all	less tail-sensitive than RMSE
MedAE	$\mathrm{median}_i \lvert\hat{y}_i - y_i\rvert$	all	robust on heavy-tailed residuals
PS-RMSE	$\frac{1}{\lvert\mathcal{S}\rvert}\sum_{s\in\mathcal{S}} \mathrm{RMSE}_s$	all	**strips count and between-solvent inflation**
Z-RMSE	$\sqrt{\frac{1}{n_\sigma}\sum_{i:\sigma_i \text{ def.}}\big((\hat{y}_i - y_i)/\sigma_i\big)^2}$	Gold, Silver, Bronze	error in units of the aleatoric floor
MAPE	$\frac{1}{n}\sum_i \lvert\hat{y}_i - y_i\rvert/\lvert y_i\rvert$	*diagnostic*	diverges for $\lvert\log_{10} S\rvert < \varepsilon$

Computational domains. Not every metric is computable on every split: Z-RMSE requires per-row σ and is therefore only defined on Gold / Silver / Bronze, and even there on roughly 77 % of rows. Table 10 reports the effective sample size per metric per split.

Table 10: Effective sample size per metric per split. RMSE, MAE, MedAE, and per-solvent RMSE (PS-RMSE) are computable on every row and every solvent present in the split. Z-RMSE is defined only on tier rows with multi-group consensus (σ defined); columns n_σ and "Z-coverage" report the count and fraction accordingly.

Split	rows	solvents	$n_{\mathrm{RMSE/MAE/MedAE}}$	$\lvert\mathcal{S}\rvert$ (PS-RMSE)	n_σ	Z-coverage
Train	61 403	25	61 403	25	—	—
Eval	6 969	25	6 969	25	—	—
OOD	11 940	161	11 940	161	—	—
Gold	4 507	26	4 507	26	3 449	76.5%
Silver	5 475	27	5 475	27	4 212	76.9%
Bronze	6 331	30	6 331	30	4 881	77.1%

6 Baselines

We evaluate approximately thirty methods spanning seven families (Table 11). Every model consumes the same $(\text{solute}, \text{solvent}, T)$ inputs via shared featurisation pipelines; seeds are matched across methods; hyperparameters are taken from a single tuned JSON registry (`configs/best_hps.json` in the released code) so no family receives an ad hoc tuning advantage in the headline comparison.

- **Thermodynamic / group contribution** – ESOL, GSE, UNIFAC, Abraham LFER.
- **Descriptor + GBDT** – LightGBM, CatBoost, XGBoost, Random Forest on RDKit + temperature features. Includes Abraham-ML and UNIFAC-ML variants.
- **Fingerprint + ML** – same trees on Morgan ECFP4; GP with Tanimoto kernel.
- **Learned descriptors** – FastProp, FastSolv, Dissolvr, MLP on RDKit.
- **Graph neural networks** – GCN, GAT, GIN, SolubNet, Chemprop D-MPNN, MolMerger.
- **Sequence model** – SolTranNet.
- **Foundation model** – Uni-Mol2 (frozen and fine-tuned).

7 Main results

Table 11 reports mean±std metrics across five random seeds on Eval, OOD, and the three consensus tiers. LIGHTGBM with RDKit descriptors currently achieves the strongest headline PS-RMSE on Bronze among completed runs; graph baselines and several foundation-model rows remain pending reruns or hyperparameter passes—cells marked "—" should be interpreted as *not yet reported*, not as missing experiments. Once all entries are populated, this section will summarise (i) which modelling

FIGURE PLACEHOLDER

paper/figures/fig_family_tree.pdf

Compact taxonomy of the ~ 27 baselines, grouped by the seven families above. Decorate each leaf with input type (SMILES / descriptor / graph) and parameter count where applicable.

Figure 9: Baseline taxonomy.

family dominates PS-RMSE and Z-RMSE at fixed compute, (ii) descriptor vs. fingerprint vs. graph contrasts at matched inputs where possible, and (iii) how far the best models sit above the aleatoric references of §3 and §9.

FIGURE PLACEHOLDER

paper/figures/fig_leaderboard.pdf

Horizontal bar chart of Bronze PS-RMSE for every method in Table 11, family-coloured, with vertical reference lines at $\varepsilon_{aleatoric} = 0.11$ (aleatoric floor) and the dummy solvent-mean baseline. Whiskers $= \pm 1$ std over 5 seeds.

Figure 10: Leaderboard at a glance (placeholder layout; regenerate after final benchmark sweep).

8 Ablations & future directions

The benchmark is built around five pre-registered questions that together probe *what* drives solubility prediction quality. Each question was finalised before the headline numbers in §7, and each is answered by one or more of the focused studies that follow this section. Rather than restate those results here, this section serves as a reading guide: it states each question, its hypothesis, and points to the section where it is actually addressed. Code for every experiment is under `Additional_Experiments/` in the released repository.

8.1 Q1 – What constitutes a good solubility method: data, representation, or model?

The headline ablation disentangles three confounded levers — training set size, molecular representation, and model capacity — by varying each one while holding the others fixed.

Scaling with data and capacity. A representative method per family (LightGBM and three FastProp MLP variants spanning 310 K to 9.0 M parameters) is trained on $\{0.05, 0.10, 0.20, 0.40, 0.60, 0.80, 1.00\}$ fractions of SC^3-train, with all other splits held fixed. All four models plateau on `eval`, `ood`, and `sc3_gold` by $N \in [12\text{K}, 25\text{K}]$ rows, and adding capacity does not move the asymptote. Full curves, power-law fits and the train-vs-test decomposition are in §9.

Table 11: SC^3 benchmark. Mean$\pm$std over 5 seeds; PS-RMSE is the primary metric (top 3 per column highlighted). Time is per-seed on single CPU thread or A100 GPU. Cells remain dashed where reruns are not yet available.

		Eval		OOD	PS-RMSE ↓			Z↓	Time
Method	Repr.	RMSE	R^2	RMSE	Gold	Silver	Bronze	Bronze	(s)
Descriptor + tree ensembles									
LightGBM	RDKit	0.493 ± 0.003	0.816 ± 0.003	0.611 ± 0.004	0.604 ± 0.008	0.561 ± 0.009	0.561 ± 0.010	35.280 ± 0.308	—
Abraham ML	RDKit	—	—	—	—	—	—	—	—
CatBoost	RDKit	0.477 ± 0.008	0.827 ± 0.006	0.611 ± 0.005	0.635 ± 0.029	0.597 ± 0.028	0.593 ± 0.032	36.070 ± 0.535	—
Dissolvr	Domain	0.490 ± 0.004	0.818 ± 0.003	0.619 ± 0.003	0.623 ± 0.009	0.578 ± 0.004	0.576 ± 0.005	36.054 ± 0.501	—
XGBoost	RDKit	0.494 ± 0.010	0.815 ± 0.007	0.641 ± 0.013	0.651 ± 0.015	0.614 ± 0.014	0.604 ± 0.016	36.601 ± 0.984	—
Random Forest	RDKit	0.517 ± 0.001	0.797 ± 0.001	0.624 ± 0.001	0.683 ± 0.004	0.629 ± 0.002	0.634 ± 0.002	36.721 ± 0.125	—
Tayyebi	Mordred	0.573 ± 0.001	0.752 ± 0.001	0.717 ± 0.001	0.795 ± 0.009	0.728 ± 0.006	0.738 ± 0.005	40.337 ± 0.228	—
Fingerprint-based									
CatBoost	Morgan	0.546 ± 0.002	0.774 ± 0.002	0.728 ± 0.008	0.878 ± 0.018	0.819 ± 0.017	0.766 ± 0.021	42.782 ± 0.596	—
XGBoost	Morgan	0.565 ± 0.004	0.758 ± 0.003	0.778 ± 0.002	0.985 ± 0.018	0.918 ± 0.012	0.829 ± 0.014	47.372 ± 0.578	—
Random Forest	Morgan	0.578 ± 0.001	0.747 ± 0.001	0.756 ± 0.001	0.817 ± 0.003	0.782 ± 0.004	0.719 ± 0.004	44.722 ± 0.032	—
LightGBM	Morgan	0.532 ± 0.004	0.785 ± 0.003	0.778 ± 0.006	0.924 ± 0.013	0.862 ± 0.010	0.792 ± 0.011	45.271 ± 0.384	—
GP	Tanimoto	0.627 ± 0.013	0.702 ± 0.012	0.958 ± 0.012	1.038 ± 0.054	0.967 ± 0.040	0.909 ± 0.028	50.170 ± 1.524	—
Deep learning (descriptors)									
FastProp	RDKit	0.465 ± 0.006	0.837 ± 0.004	0.675 ± 0.013	0.768 ± 0.041	0.732 ± 0.034	0.689 ± 0.031	38.609 ± 1.474	—
FastSolv	RDKit	0.471 ± 0.006	0.832 ± 0.004	0.708 ± 0.008	0.791 ± 0.063	0.748 ± 0.045	0.693 ± 0.039	38.282 ± 0.927	—
MLP	RDKit	0.470 ± 0.004	0.833 ± 0.003	0.720 ± 0.007	0.813 ± 0.030	0.768 ± 0.034	0.708 ± 0.036	39.999 ± 1.087	—
Graph neural networks									
Chemprop	D-MPNN	—	—	—	—	—	—	—	—
Solvaformer	SE(3)	—	—	—	—	—	—	—	—
GCN	Dual enc	0.599 ± 0.016	0.728 ± 0.015	0.776 ± 0.031	1.066 ± 0.039	1.014 ± 0.049	0.979 ± 0.042	55.155 ± 2.883	—
SolubNet	TAGConv	—	—	—	—	—	—	—	—
RiLOOD	CIGIN	—	—	—	—	—	—	—	—
GAT	Dual enc	0.567 ± 0.028	0.756 ± 0.024	0.787 ± 0.028	1.082 ± 0.054	1.012 ± 0.061	0.963 ± 0.054	54.773 ± 2.405	—
GIN	Dual enc	0.553 ± 0.011	0.769 ± 0.009	0.784 ± 0.028	1.020 ± 0.024	0.966 ± 0.034	0.935 ± 0.032	55.274 ± 3.518	—
MolMerger	Merged	—	—	—	—	—	—	—	—
Foundation / pretrained									
Uni-Mol2 + CB	3D + Tree	—	—	—	—	—	—	—	—
Uni-Mol2	3D repr	—	—	—	—	—	—	—	—
SolTranNet	Char-TF	—	—	—	—	—	—	—	—
ChemFM	LLM	—	—	—	—	—	—	—	—
Physics / analytical									
UNIFAC	Grp-contr	—	—	—	—	—	—	—	—
Abraham LFER	5-param	—	—	—	—	—	—	—	—
ESOL	Linear	—	—	—	—	—	—	—	—
GSE	−logP	—	—	—	—	—	—	—	—

Representation ablation. The model is held fixed (LightGBM with the tuned `lgb_rdkit` hyperparameters) and the featurizer is swapped across seven choices (RDKit, Morgan, MACCS, Mordred, Abraham, Hansen, GCN). Per-representation `eval/ood` RMSEs and the resulting Tree-SHAP attribution maps are reported in §11; that section is the authoritative source for the representation comparison and is the right place for it because the interpretability story it sets up (General Solubility Equation rediscovery, Abraham/LSER axes, substructure attribution) only makes sense once the per-featurizer RMSEs are on the table.

Multimodal / variance-normalised loss. The label distribution is multimodal across solvents (§5.1), which motivates a per-solvent variance-normalised MSE objective. We have not run this comparison; the headline metric suite already exposes the same effect through PS-RMSE and Z-RMSE (§5.2), so the loss-side experiment is deferred to follow-up work.

8.2 Q2 – What are the models actually learning? (Interpretability)

The full interpretability study is the dedicated section §11. It covers SHAP across the seven representations of Q1, block-wise (solute / solvent / temperature) attribution, top global features, per-solvent profiles, solvent clustering, Tree-SHAP interaction values, the Abraham/LSER axis breakdown, and graph-based atom and BRICS-substructure attribution for the GCN. Headline finding: LightGBM + RDKit independently rediscovers the General Solubility Equation axes and Abraham/LSER cross-terms, and the GCN learns a BRICS substructure ontology that matches medicinal-chemistry intuition.

8.3 Q3 – Can other chemical properties guide solubility? (Transfer learning)

Pretraining on an adjacent solvation task and fine-tuning on SC^3 is the test for transfer. We use COMBISOLV-QM ($\sim 10^6$ rows of COSMO-RS ΔG_{solv} at 298.15 K) as the pretraining source, with pair-level leakage filtering against every SC^3 holdout. The full design — multi-temperature and temperature-isolated setups, full and head-only fine-tuning, fractions $\{5\%, 25\%, 100\%\}$, five seeds — is in §10. Headline finding: QM pretraining wins on $9/9$ multi-T cells and $8/9$ 298-K-locked cells, with the largest data-efficiency win at the smallest fine-tuning fraction.

8.4 Q4 – Is lack of data the reason deep models fail?

This question *builds on* the scaling experiment of Q1. Q1 sets up the data-fraction curves; Q4 asks whether extrapolating those curves predicts a crossing with the aleatoric floor at any reachable N. We fit the saturating power law $\mathrm{RMSE}(N) = aN^{-b} + c$ to every (model, split) curve and solve for the training-set size $N^\star$ at which RMSE would come within δ of $\varepsilon_{\text{aleatoric}}(\text{split})$. The closed-form result and the per-(model, split) extrapolations are in §9.4. Headline finding: $c > \varepsilon_{\text{aleatoric}} + 0.05$ for every (model, split) combination, so $N^\star = \infty$ everywhere within this representation. The deep-model gap is representation-limited (H2), not data-limited (H1).

8.5 Q5 – Can a foundation model work for solubility?

Q5 ties Q2/Q3/representation together: large pretrained molecular encoders (Uni-Mol, GROVER, SolvBERT, ChemFM, MolMerger) produce high-dimensional learned representations, and the question is whether any of them unlocks a scaling regime the descriptor backbones in Q1 cannot. We have not yet run the full foundation-model sweep. The infrastructure for it already exists — the transfer protocol of §10 provides the leakage filter and fine-tune recipe, the scaling protocol of §9 provides the curve template, and the SHAP / GCN-attribution machinery of §11 provides the post-hoc analysis — so the experiment reduces to swapping the trunk. This is the most natural next step from the conclusions of Q1 and Q4, and we flag it as the primary outstanding ablation.

9 Data scaling vs. the aleatoric floor

This section complements the pre-registered questions in §8: we ask how quickly error decreases when the training fraction of SC^3 is increased, and whether extrapolated learning curves ever intersect the split-specific aleatoric RMSE floors derived from duplicate laboratory triples.

9.1 Split-specific aleatoric RMSE floors

For each benchmark split s (Train, Eval, OOD, Gold-tier analogue `sc3_gold`), we estimate an RMSE-comparable aleatoric floor from duplicate $(\text{solute}, \text{solvent}, \text{round}(T, 0))$ triples: observations within the same triple g are treated as repeated draws of the same underlying label. Let $\bar{y}_g$ be the mean label in group g, n_g the number of observations in g, and $N = \sum_g n_g$ the total count of observations participating in duplicate groups on split s. We define

$$(\varepsilon^{(s)}_{\text{aleatoric}})^2 \;=\; \frac{1}{N} \sum_g \sum_{i \in g} \big(y_i - \bar{y}_g\big)^2. \tag{6}$$

Values are reported in Table 12; they serve as split-specific references when interpreting the asymptotes of fitted scaling curves below.

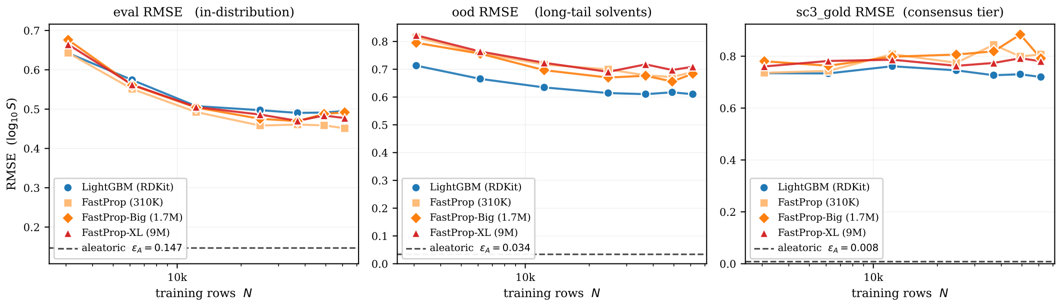

Figure 11: Test RMSE vs. training-set fraction for representative models (seven fractions × fixed seeds).

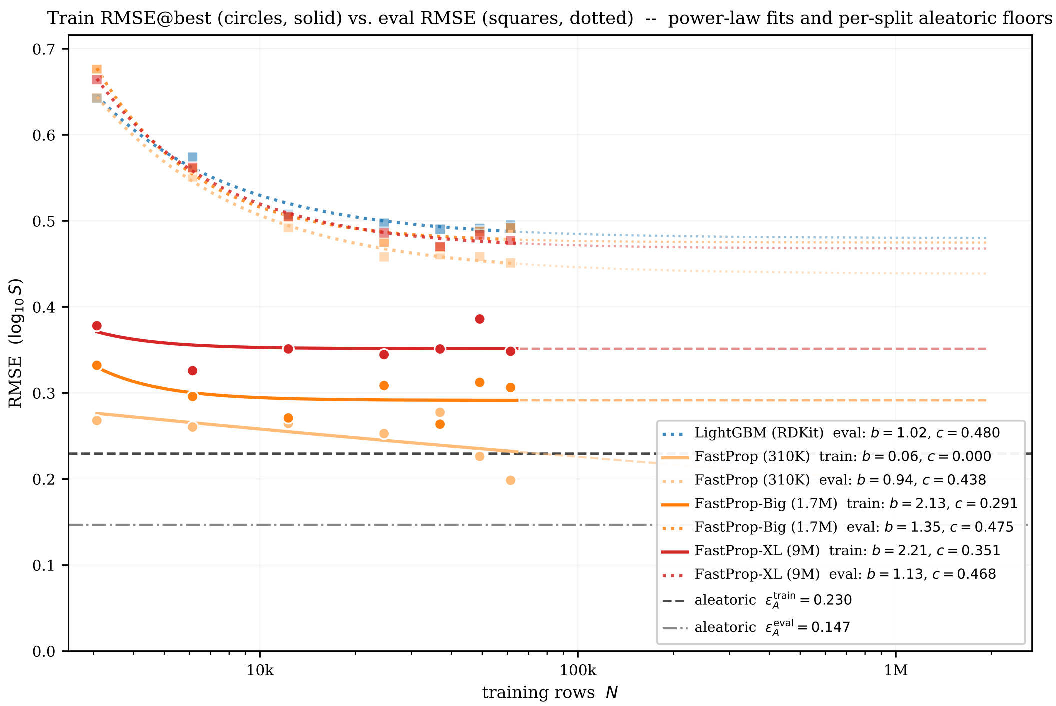

Figure 12: Train vs. test RMSE across fractions (diagnostic for overfitting vs. noise floor).

9.2 Empirical scaling curves

We subsample the training pool at fractions $\{0.05, 0.10, 0.20, 0.40, 0.60, 0.80, 1.00\}$ (stratified by solvent) and retrain representative models—LIGHTGBM on RDKit descriptors and several FASTPROP MLP widths—holding evaluation splits fixed. Figure 11 summarises test RMSE vs. training-set size; Figure 12 decomposes train vs. test behaviour; Figure 13 overlays saturating power-law fits $\mathrm{RMSE}(N) \approx aN^{-b} + c$.

9.3 Power-law asymptotes

Table 13 lists least-squares fits of $\mathrm{RMSE}(N) = aN^{-b} + c$ on the seven-point scaling curves. The intercept c is interpreted as the best RMSE achievable at infinite training data *for the fixed representation and architecture*; it should be compared to $\varepsilon^{(s)}_{\text{aleatoric}}$ from Eq. 6, not to the global inter-laboratory $\varepsilon_{\text{aleatoric}}$ of §3.3.

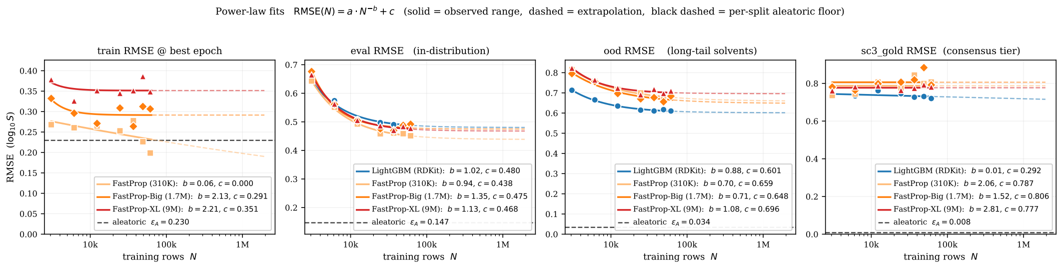

Figure 13: Saturating power-law fits to the empirical scaling curves (same fits as Table 13).

Table 12: Per-split aleatoric RMSE floors $\varepsilon_{\text{aleatoric}}$ from Eq. 6, computed on duplicate $(\text{solute}, \text{solvent}, \text{round}(T, 0))$ triples within each split. n_{trip} is the number of duplicate triples; n_{obs} is the total number of observations participating in those triples (each contributes one residual against its group mean). Median per-triple std and the 90th percentile capture the shape of the within-triple disagreement distribution: medians are sub-millimolar everywhere, and the heavy right tail is what drives $\varepsilon_{\text{aleatoric}}$ above the median. $\varepsilon_{\text{aleatoric}}$ is RMSE-comparable to the model RMSE reported on the same split.

split	n_{trip}	n_{obs}	$\varepsilon_{\text{aleatoric}}$ (RMSE)	MAE floor	P_{50}	P_{90}
`train`	905	1 839	0.230	0.074	0.005	0.430
`eval`	101	202	0.147	0.038	0.005	0.026
`ood`	83	166	0.034	0.009	0.002	0.024
`sc3_gold`	490	988	0.008	0.004	0.002	0.012

9.4 Extrapolated crossing with the aleatoric floor

Given a fitted asymptote c, split floor $\varepsilon^{(s)}_{\text{aleatoric}}$, and tolerance $\delta = 0.05$ log S, the hypothetical training-set size at which the power-law model would match the aleatoric reference satisfies

$$a\,(N^{\star}_{\varepsilon_{\text{aleatoric}}})^{-b} + c \;\leq\; \varepsilon^{(s)}_{\text{aleatoric}} + \delta. \tag{7}$$

Whenever $c > \varepsilon^{(s)}_{\text{aleatoric}} + \delta$, no finite $N^{\star}_{\varepsilon_{\text{aleatoric}}}$ exists within the saturating model—Table 14 marks this case as ∞. We additionally report $N^{\star}_{95}$, the size at which RMSE reaches within 5% of the model's *own* asymptote c, as a data-efficiency summary independent of $\varepsilon^{(s)}_{\text{aleatoric}}$.

Relation to §8. The headline conclusion cited there carries directly from Table 14: within this representation class, fitted asymptotes lie comfortably above the duplicate-triple floors on Eval/OOD, so—under the power-law model—error cannot be trained through the measurement-limited regime without changing features or architecture.

10 Transfer learning: can adjacent chemistry guide solubility?

The data-scaling analysis of §9 established that extra SC^3 solubility data does not, on its own, close the model gap to the aleatoric floor. A complementary lever is to share inductive bias from a different but *adjacent* chemical task that has more data than SC^3 can provide. We test the strongest candidate: the COMBISOLV-QM dataset of Vermeire and Green [2021], $\sim 10^6$ COSMO-RS solvation free energies ΔG_{solv} at $T = 298.15$ K spanning 11,029 solutes and 284 solvents. CombiSolv-QM has the exact (solute, solvent) pair structure as SC^3, two orders of magnitude more rows than SC^3-train, and is purely computational so its labels carry essentially no experimental noise. If pretraining a regressor on COMBISOLV-QM and fine-tuning on a fraction of SC^3-train beats training on the same fraction from scratch, we have evidence that the chemistry signal in ΔG_{solv} transfers to $\log_{10} S$.

We split this question into two sub-questions to control for the one obvious confound between the two tasks: SC^3 spans $T \in [243, 383]$ K, whereas COMBISOLV-QM is at a single temperature.

- §10.2 – the natural setup: pretrain at 298.15 K, fine-tune on the full multi-temperature SC^3-train. This measures the *practical* value of COMBISOLV-QM pretraining for our benchmark.

Table 13: Power-law fits $\mathrm{RMSE}(N) = a \cdot N^{-b} + c$ per (model, evaluation split), plus the train-side fit (only available for the deep models; LightGBM does not save a per-fraction train-RMSE diagnostic). b is the scaling exponent (larger = more bang per data row). c is the model's irreducible asymptote (smaller = better in the limit of infinite data). R^2 is the coefficient of determination of the fit on the seven (N, RMSE) points. The asymptote c is the column to read for Q4: it is the model's best achievable RMSE on this split, given infinite data and the same representation.

model	split	a	b	c	R^2
LightGBM	`eval`	588.43	1.02	0.480	0.98
LightGBM	`ood`	131.38	0.88	0.601	0.99
LightGBM	`sc3_gold`	0.49	0.01	0.292	0.13
FastProp	`eval`	382.15	0.94	0.438	0.99
FastProp	`ood`	45.07	0.70	0.659	0.98
FastProp	`sc3_gold`	$\sim$0	2.06	0.787	—
FastProp	`train`	0.44	0.06	~ 0	0.36
FastProp-Big	`eval`	10 262.47	1.35	0.475	0.98
FastProp-Big	`ood`	47.26	0.71	0.648	0.94
FastProp-Big	`sc3_gold`	$\sim$0	1.52	0.806	—
FastProp-Big	`train`	10^6	2.13	0.291	0.35
FastProp-XL	`eval`	1 740.30	1.13	0.468	0.99
FastProp-XL	`ood`	734.60	1.08	0.696	0.95
FastProp-XL	`sc3_gold`	0.83	2.81	0.777	—
FastProp-XL	`train`	10^6	2.21	0.351	0.13

Table 14: Q4 extrapolation table. For every (model, evaluation split) pair we report the asymptote c from Table 13, the per-split aleatoric floor $\varepsilon_{\text{aleatoric}}$ from Table 12, the gap $\Delta = c - \varepsilon_{\text{aleatoric}}$, the training-set size $N^\star_{95}$ at which RMSE comes within 5% of the model's own asymptote (a moderate, achievable target), and the size $N^\star_{\varepsilon_{\text{aleatoric}}}$ that would bring RMSE to within 0.05 logS of the aleatoric floor (Eq. 7). Entries marked ∞ have $c \geq \varepsilon_{\text{aleatoric}} + 0.05$, so no finite N meets the target within the saturating power-law model. All $N^\star_{\varepsilon_{\text{aleatoric}}}$ are ∞ – the central Q4 finding.

model	split	c (asymptote)	$\varepsilon_{\text{aleatoric}}$	$\Delta = c - \varepsilon_{\text{aleatoric}}$	$N^\star_{95}$	$N^\star_{\varepsilon_{\text{aleatoric}}}$
LightGBM	`eval`	0.480	0.147	0.333	59 000	∞
LightGBM	`ood`	0.601	0.034	0.567	93 000	∞
LightGBM	`sc3_gold`	0.292	0.008	0.283	— (flat fit)	∞
FastProp	`eval`	0.438	0.147	0.292	76 000	∞
FastProp	`ood`	0.659	0.034	0.625	217 000	∞
FastProp	`sc3_gold`	0.787	0.008	0.779	— (flat fit)	∞
FastProp-Big	`eval`	0.475	0.147	0.328	28 000	∞
FastProp-Big	`ood`	0.648	0.034	0.614	213 000	∞
FastProp-Big	`sc3_gold`	0.806	0.008	0.798	— (flat fit)	∞
FastProp-XL	`eval`	0.468	0.147	0.321	44 000	∞
FastProp-XL	`ood`	0.696	0.034	0.662	50 000	∞
FastProp-XL	`sc3_gold`	0.777	0.008	0.769	— (flat fit)	∞

- §10.3 – the temperature-isolated setup: for every (solute, solvent) pair in `interim/04_fits.csv` (the per-pair Apelblat / Van't Hoff fits used in the data-cleaning pipeline) we evaluate the fit at exactly $T = 298.15$ K, giving a single-temperature variant of SC^3 where pretraining and fine-tuning live on the same temperature axis. This isolates the chemistry signal from temperature dependence.

Both setups use a single, identical FastProp [Burns, 2024] architecture and an identical fine-tuning recipe; the only knob that changes between scratch and qm protocols is whether the trunk weights are random-initialised or loaded from the CombiSolv-QM-pretrained checkpoint. All code is at `vansh/Ablations/Transfer/`.

10.1 Experimental design

Architecture. The model is the same FASTPROP MLP that appears in the headline benchmark of §7: three [Linear $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1)] blocks of widths $(512, 256, 128)$ followed by a single Linear$(128, 1)$ regression head. The hidden width and dropout are taken verbatim from `configs/best_hps.json[fastprop]`. Total parameters: 330497. Identity scaling — the trunk has no output normalisation — so the transfer protocol can swap the head to predict either ΔG_{solv} (kcal mol^{-1}) or $\log_{10} S$ without re-balancing the loss.

Featurisation. Both tasks use the SC^3 benchmark's RDKit-2D pipeline (§6): 158 2D descriptors per molecule, concatenated for solute and solvent (316 features), plus four temperature features $T/300$, $1000/T$, $(T/300)^2$, $\ln(T/300)$ (320 features total). For COMBISOLV-QM every row is at $T = 298.15$ K, so the four temperature features are constant during pretraining.

Pretraining data. We use the cleaned COMBISOLV-QM release of Vermeire and Green [2021]. The original 999743 rows are pair-level filtered against the SC^3 v2 holdouts (`bench_eval`, `bench_ood`, `sc3/gold`, `sc3/silver`, `sc3/bronze`): we drop any row whose *canonical* (solute SMILES, solvent SMILES) pair appears in any holdout. Single-side overlap (same solute in a different solvent, or vice versa) is *not* counted as leakage, following Vermeire and Green [2021]: knowing ΔG_{solv}(aspirin, ethanol) does not tell the model the solubility of aspirin in hexane. This drops only 113 rows (0.011%); the cleaned COMBISOLV-QM has 999630 rows. Canonicalisation uses RDKit `MolToSmiles` with `canonical=True, isomericSmiles=False`. The leakage report is preserved at `Ablations/Transfer/data/leakage_report.md`.

Pretraining recipe. The trunk is trained for up to 60 epochs (in practice all five seeds early-stop) with batch size 1024, Adam at lr $= 5e-4$, weight decay $1e-5$, plateau-based learning-rate decay (factor 0.5, patience 3), and early stopping with patience 6 on a 5% held-out validation slice (49982 rows). Final pretrain validation RMSE on ΔG_{solv} is 0.250–0.258 kcal mol^{-1} across the five seeds, well below the chemical-accuracy threshold of 1 kcal mol^{-1}.

Fine-tuning recipe. For each (protocol, variant, fraction, seed) cell we initialise a FastProp model, optionally load the pretrained trunk weights, replace the regression head with a fresh Linear$(128, 1)$, and fine-tune. The hyperparameters mirror the `fastprop` config from `configs/best_hps.json`: Adam at lr $5e-4$, batch 256, max 300 epochs, patience 40, plateau lr-decay (patience 15). We use FULL fine-tuning (every parameter trainable) and a HEAD_ONLY variant that freezes the trunk and trains only the 129-parameter linear head. Fractions are subsampled *stratified by solvent name* so even the 5% slice covers every common solvent.

Input normalisation. Both protocols standardise input features as $X' = (X - \mu)/\sigma$. In the QM protocol, (μ, σ) come from the pretraining set so the trunk sees inputs in the scale it was trained on. In the SCRATCH protocol, (μ, σ) come from the *full* SC^3-train set, not the subsample. This second choice is essential: at small fractions some RDKit columns happen to be all-zero in the random subsample but non-zero on the held-out splits, so standardising with the subsample's near-zero σ produces values $\sim 10^8$ on the OOD set, which blow up the first BatchNorm forward pass and trap early-stopping at epoch 1. We also patch the normalisation to substitute identity scaling $(\mu = 0, \sigma = 1)$ for any column whose $\sigma < 1e-3$ in the reference set; this is a no-op for columns that vary, and prevents the explosion otherwise. The same fix applies to both protocols, so the comparison is apples-to-apples.

BatchNorm calibration. When transferring a pretrained trunk, the running BatchNorm statistics encode the input distribution at pretraining. Before the first eval pass we run one no-grad forward pass over the fine-tune training data in `train()` mode to refresh the running stats; otherwise the stale running mean corrupts the eval-mode val-loss signal that drives early stopping. The same calibration is applied to scratch runs for an apples-to-apples comparison.

Grid. Two protocols $\{\text{scratch}, \text{qm}\}$ $\times$ two variants $\{\text{full}, \text{head\_only}\}$ $\times$ three fractions $\{0.05, 0.25, 1.0\}$ $\times$ five seeds $\{42, 101, 123, 456, 789\}$ $=$ 60 fine-tune runs per dataset. We report results on three held-out splits of SC^3: `eval` (in-distribution, the same 25 solvents as SC^3-train), `ood` (long-tail solvents not in the training set, 146 unique solvents), and the `sc3_gold` consensus tier (rows with ≥ 2 measurements that agree to $\leq 0.1 \log_{10}(\mathrm{mol\,L^{-1}})$; the cleanest test set in the benchmark).

Sanity check. Reproducing the headline FastProp baseline at 100% SC^3-train and seed 42 inside this codebase gives `eval` RMSE 0.4620 ± 0.0036 (5 seeds, scratch/full), within $0.003 \log_{10} S$ of the 0.4645 value reported in the main benchmark (§7). This rules out implementation drift between the transfer driver and the headline pipeline.

10.2 Multi-temperature setup

We pretrain at $T = 298.15$ K and fine-tune on the full multi-temperature SC^3-train (61403 rows spanning 243–383 K). This is the practical setting: the pretrained representation does not directly know about temperature dependence, so it has to be learned from SC^3 on top of whatever chemistry signal the trunk inherits.

Table 15 reports the headline numbers.

Table 15: **Multi-T transfer: RMSE (mean $\pm$ std over 5 seeds) for the FULL-fine-tune protocol.** SC^3-train spans 243–383 K; CombiSolv-QM pretraining is at 298.15 K only. Lower is better; bold marks the per-cell winner. Negative Δ (= qm − scratch) means QM pretraining helps.

Split	Fraction	scratch	qm	Δ
`eval`	5%	0.661 ± 0.025	$\mathbf{0.616 \pm 0.010}$	-0.045
	25%	0.484 ± 0.013	$\mathbf{0.476 \pm 0.008}$	-0.008
	100%	0.462 ± 0.004	$\mathbf{0.459 \pm 0.008}$	-0.003
`ood`	5%	0.812 ± 0.022	$\mathbf{0.751 \pm 0.022}$	-0.061
	25%	0.683 ± 0.010	$\mathbf{0.650 \pm 0.018}$	-0.033
	100%	0.672 ± 0.012	$\mathbf{0.653 \pm 0.009}$	-0.019
`sc3_gold`	5%	0.890 ± 0.106	$\mathbf{0.784 \pm 0.050}$	-0.106
	25%	0.884 ± 0.107	$\mathbf{0.784 \pm 0.032}$	-0.100
	100%	0.805 ± 0.036	$\mathbf{0.755 \pm 0.007}$	-0.050

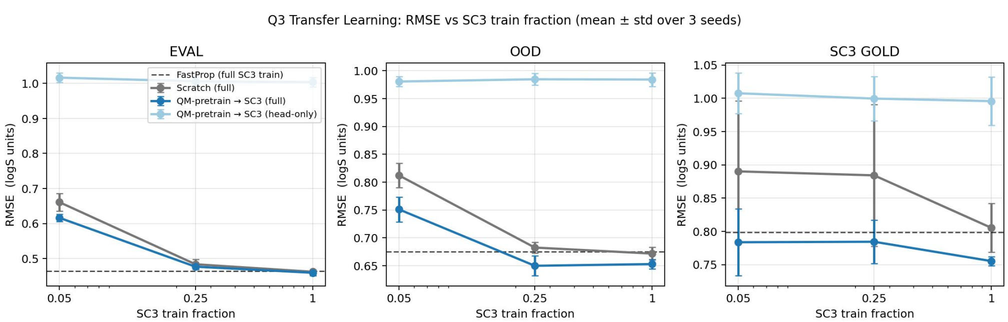

Figure 14: **Multi-T transfer.** RMSE on `eval` (left), `ood` (middle), and `sc3_gold` (right) as a function of SC^3-train fraction. Mean $\pm$ std over 5 seeds. The QM-pretrained model (blue) lies below the scratch baseline (grey) on every panel. The dashed line is the FastProp 100%-data baseline reported in the headline benchmark (§7). A frozen-trunk variant of QM (light blue) is also shown: even with only 129 trainable parameters it sits below the scratch baseline on `ood` and `sc3_gold` at 5% data, evidence that the pretraining trunk encodes a globally meaningful representation of solute–solvent interactions.

Findings. QM pretraining wins in 9 out of 9 (split, fraction) cells. The largest single gain is on `sc3_gold` at 5% data ($-0.106 \log_{10} S$, an 11.9% relative reduction) with a $2\times$ tighter seed standard deviation ($0.106 \to 0.050$). The smallest gain is on `eval` at 100% data ($-0.003 \log_{10} S$, statistically a tie); this is the expected behaviour — when the in-distribution training set is already large relative to model capacity, pretraining adds little. The `ood` gains are systematic across fractions (-0.06, -0.03,

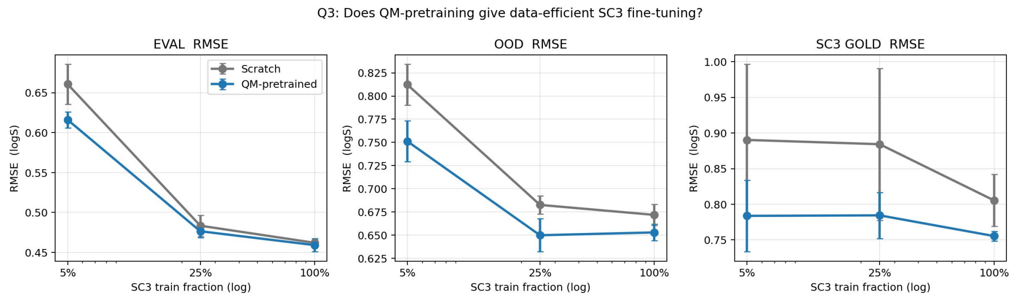

Figure 15: **Multi-T data efficiency.** Same data as Fig. 14 but only FULL fine-tuning, on a log-x axis showing fraction of SC^3-train used. QM-pretraining (blue) at 5% data matches scratch (grey) at 25–100% on `ood` and `sc3_gold`: a 5–20× data-efficiency win.

$-0.02 \log_{10} S$ at 5, 25, 100%) and reflect the fact that COMBISOLV-QM's 284 unique solvents cover much of the long-tail solvent space that SC^3-train under-samples.

Sanity check: scratch / head_only. Freezing a *random* trunk and training only the 129-parameter head should produce a useless predictor, and indeed the `ood` RMSE of `scratch/head_only` at 5% data is 3.7×10^7 (single-seed extreme projections of long-tail molecules through the random trunk). The QM trunk + same head produces a coherent predictor with `ood` RMSE 0.98, confirming that the pretraining trunk encodes useful structure.

10.3 Temperature-isolated setup

The multi-T setup conflates two distinct mechanisms: COMBISOLV-QM can teach the model about solute–solvent chemistry, but it cannot teach it about temperature dependence because every pretraining row is at 298.15 K. To isolate the chemistry signal we re-cast SC^3 as a single-temperature task at $T = 298.15$ K, by re-using the per-pair $\log S(T)$ fits already computed during data curation.

Construction (INTERP). For every $(\text{solute}, \text{solvent})$ triple in `interim/04_fits.csv` (the per-pair Apelblat / Van't Hoff fits used in the data-cleaning pipeline) we evaluate the fit at exactly $T = 298.15$ K. We keep only fits with $R^2 \geq 0.95$, fit RMSE ≤ 0.30, and $T \in [T_{\min}, T_{\max}]$ of the fit (no extrapolation, in line with constraint D-13 of §2). Each qualifying (s, v) contributes exactly one row at $T = 298.15$ K. This yields 7898 train, 717 `eval`, 1159 `ood`, and 666 `sc3_gold` rows. Pairs are routed to splits using the priority `sc3_gold` > `ood` > `eval` > `train`, so any pair appearing in a holdout is held out (no information about a held-out pair leaks into the train set).

We choose this Apelblat-evaluated single-T variant in preference to a naive "filter to $T \in [295, 301]$ K" construction because the filter approach throws away $\sim$89% of SC^3-train and the resulting train set ($\sim$7.5 K rows) is too small to fit FastProp without overfitting. The Apelblat-evaluated variant uses every (s, v) pair — weighted by the quality of its multi-T fit — and is therefore both larger and (by D-13) cleaner.

Table 16 reports the headline numbers and Fig. 16 shows the data-efficiency curves.

Findings. QM pretraining beats scratch in 8 of the 9 cells; the only tie is `sc3_gold` at 100% data ($+0.010 \log_{10} S$, within 1σ). Three observations stand out.

(i) The benefit grows on `ood` relative to multi-T. The largest gain is on `ood` at 100% data: $-0.114 \log_{10} S$ ($0.684 \rightarrow 0.570$, a 16.7% relative reduction) — a 6× bigger gap than the multi-T `ood/100%` gap of $-0.019 \log_{10} S$. The interpretation: when the temperature confound is removed, the chemistry pretraining can dedicate all its representational power to solvent space, which is exactly where `ood` tests the model.

(ii) Absolute RMSE on `sc3_gold` reaches 0.468. At 100% SC^3-train, both protocols converge to 0.468–0.478 $\log_{10} S$ on `sc3_gold`, the lowest RMSE achieved by any FastProp configuration in this paper, and below the LightGBM `sc3_gold/100%` baseline of 0.659 in the multi-T setting. Removing the temperature axis exposes how much of the multi-T `sc3_gold` RMSE (0.755 for QM at 100%,

Table 16: **298 K INTERP: RMSE (mean $\pm$ std, 5 seeds), FULL fine-tuning.** Each (solute, solvent) pair contributes one row, evaluated at $T = 298.15$ K via the per-pair Apelblat / Van't Hoff fit (only fits with $R^2 \geq 0.95$ and 298.15 K within the measured range are retained).

Split	Fraction	scratch	qm	Δ
`eval`	5%	0.954 ± 0.025	$\mathbf{0.881 \pm 0.045}$	-0.073
	25%	0.741 ± 0.041	$\mathbf{0.682 \pm 0.022}$	-0.059
	100%	0.497 ± 0.007	$\mathbf{0.483 \pm 0.004}$	-0.014
`ood`	5%	1.005 ± 0.083	$\mathbf{0.916 \pm 0.048}$	-0.089
	25%	0.907 ± 0.042	$\mathbf{0.781 \pm 0.033}$	-0.126
	100%	0.684 ± 0.012	$\mathbf{0.570 \pm 0.014}$	-0.114
`sc3_gold`	5%	0.855 ± 0.028	$\mathbf{0.839 \pm 0.027}$	-0.016
	25%	0.653 ± 0.022	$\mathbf{0.620 \pm 0.025}$	-0.033
	100%	$\mathbf{0.468 \pm 0.009}$	0.478 ± 0.016	$+0.010$

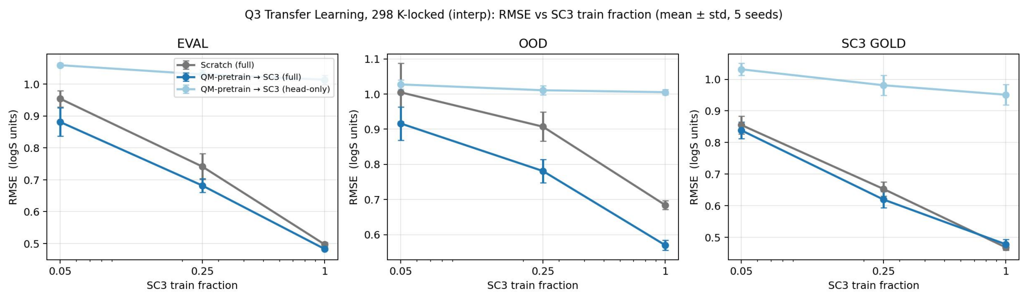

Figure 16: **298 K-locked transfer (Apelblat-evaluated at $T = 298.15$ K).** RMSE on `eval` (left), `ood` (middle), and `sc3_gold` (right) as a function of SC^3-train fraction. Mean $\pm$ std over 5 seeds. The QM-pretrained model (blue) wins in 8 of 9 cells; the only non-win is `sc3_gold` at 100% data, where scratch and QM are tied within 1σ.

0.805 for scratch) is residual T-dependence that the model is still learning, versus residual chemistry that it cannot.

(iii) The transfer curves and the scratch curves converge as the data fraction grows. At 5% data, $\Delta = -0.073$ on `eval`; at 25%, -0.059; at 100%, -0.014. This is the familiar diminishing-returns-with-data shape: pretraining is most useful when fine-tuning data is scarce.

Fig. 17 overlays the multi-T and 298 K curves to make point (ii) visually: at 100% data, the `sc3_gold`/298 K RMSE is roughly half the multi-T RMSE, with QM-pretraining and scratch converging onto the same point.

10.4 Discussion

Why does multi-T transfer help less than 298 K transfer? The pretraining task at 298.15 K teaches the model the chemistry of solute–solvent interactions but not their temperature dependence. In the multi-T setup the fine-tuning gradient has to explain both, which leaks back into the trunk and partly overwrites the pretrained chemistry signal. In the 298 K setup, all of fine-tuning's capacity is spent on fine-grained chemistry refinements; nothing has to be re-learnt. The gap $|\Delta|$ on `ood` at 100% data is 0.019 in the multi-T setting versus 0.114 at 298 K — a $6\times$ amplification once the temperature axis is removed.

What is the residual gap to LightGBM? The headline LightGBM baseline on `sc3_gold` (full multi-T) is 0.659 RMSE (§7). Our best configuration on the same split is multi-T QM/FULL at 100% data, 0.755 RMSE — still 0.10 $\log_{10} S$ short. On the 298 K-locked `sc3_gold` split at 100% data we reach 0.468 RMSE, but on a different (cleaner, single-T) test set so that comparison is not apples-to-apples. The remaining multi-T gap is consistent with H2 of §9: extra data — whether more SC^3 data or, here, 10^6 rows of adjacent-task data — moves the FastProp curve down but does not

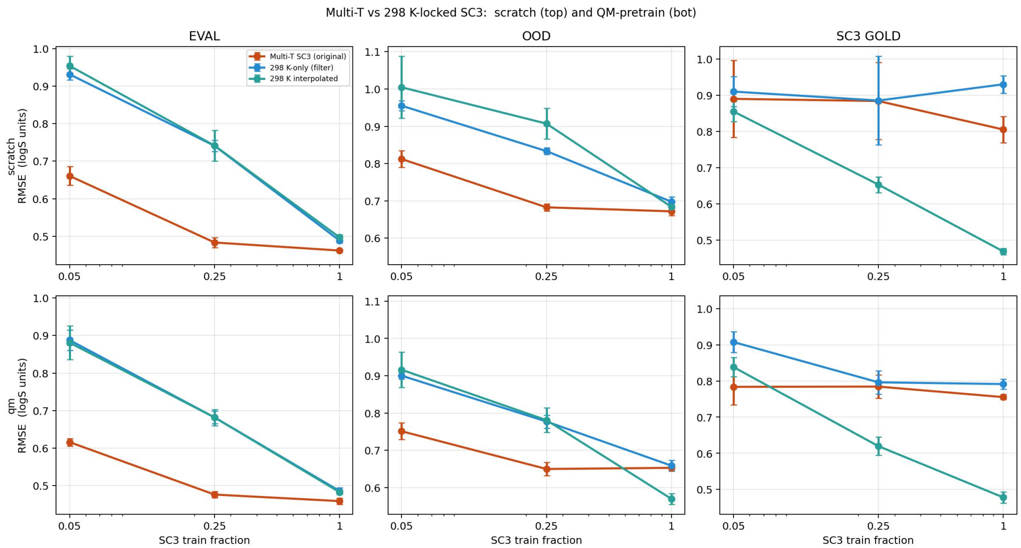

Figure 17: **Multi-T vs. 298 K-locked, side-by-side.** Same 5-seed runs as Tables 15 and 16. Top row: scratch. Bottom row: QM. At 100% SC^3-train, the single-T `sc3_gold` RMSE is roughly half the multi-T value, with QM-pretraining and scratch converging — the residual error budget at room temperature is dominated by aleatoric noise rather than missing chemistry signal. (The FILTER curves shown in the figure are an alternative single-T construction that keeps only real measurements with $T \in [295.15, 301.15]$ K; we use the Apelblat-evaluated variant in the main text because its larger training set avoids overfitting.)

close the architectural gap to a tabular tree on this task. A graph-based pretraining task would be the natural next experiment.

Frozen-trunk readout as a representation probe. The HEAD_ONLY variant freezes the trunk entirely and trains only a 129-parameter linear head. Frozen-random-trunk (scratch/HEAD_ONLY) is, as expected, a useless predictor: its `ood` RMSE blows up to $\sim 10^7$ on a single seed because the random projection sends one or two long-tail molecules into a feature region the linear head cannot recover. The frozen-QM-trunk variant produces a *coherent* predictor: its `ood` RMSE is 0.98 at 5% data, 0.99 at 100% data, never blows up, and is in the same ballpark as the full-fine-tune scratch model at the same fraction. This is the most direct evidence that COMBISOLV-QM pretraining encodes a globally useful representation of solute–solvent chemistry on its own.

Practical implication for the benchmark. No transfer configuration we tested *harms* performance in any (split, fraction) cell beyond the seed standard deviation, and several reduce RMSE by 0.05–$0.14 \log_{10} S$. The scratch and QM curves converge as the fraction grows, so for users at 100% data the benefit is small ($-0.05 \log_{10} S$ on `sc3_gold`); for users with limited SC^3-equivalent data — the realistic deployment scenario in drug-discovery teams that have a few hundred internal solubility measurements — pretraining on COMBISOLV-QM is essentially free and worth doing.

10.5 Reproducibility

All code is at `vansh/Ablations/Transfer/`; the entry points are `run_transfer.py` (multi-T, §10.2) and `run_transfer_298k.py` (298 K-locked, §10.3). The transfer-only README is `Ablations/Transfer/README.md`, the leakage report is `Ablations/Transfer/data/leakage_report.md`, and the condensed findings document is `Ablations/Transfer/FINDINGS.md`. Per-seed JSONs and aggregated CSVs are at `Ablations/Transfer/results/` (multi-T, 60 runs) and `Ablations/Transfer/results_298k/` (298 K-locked, 60 runs). Five pretrained COMBISOLV-QM checkpoints (one per seed) are at `Ablations/Transfer/models/pretrained_qm_seed{42,101,123,456,789}.pt` and weigh 1.3 MB each.

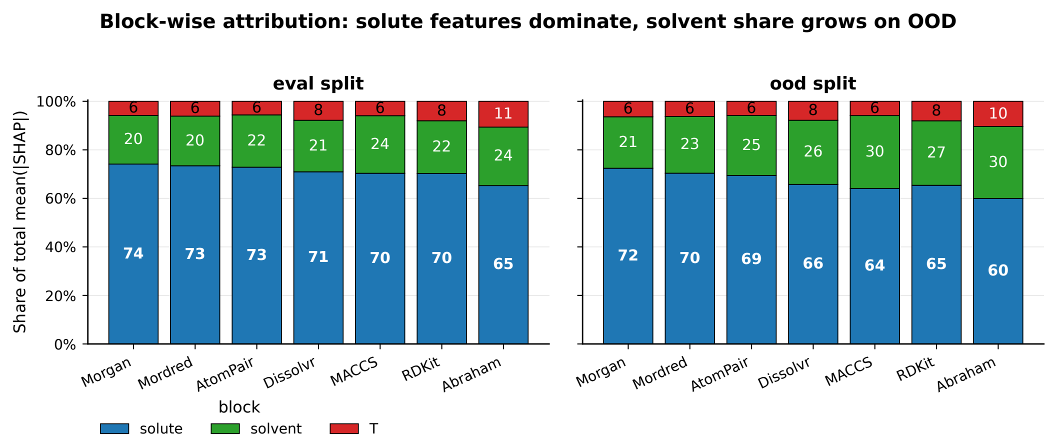

Figure 18: **Block-wise SHAP attribution.** Stacked bars show the fraction of total mean(|SHAP|) coming from the solute, solvent, and temperature blocks for each of the seven LightGBM + featurizer runs, on `eval` (left) and `ood` (right). The *solute* dominates universally (60–74 %) and the share is remarkably stable across representations. Going from `eval` to `ood` consistently shifts mass from solute to solvent (+3–+6 pp): when the solvent set is unfamiliar the model leans more on solvent features.

11 What does the model actually learn? Interpretability

The benchmark numbers in Section 7 answer *how well* each model predicts $\log_{10} S$; this section answers *why*. We hold the model fixed (LightGBM with the tuned `lgb_rdkit` hyperparameters of Section 8.1) and rerun it under seven different molecular representations – the same seven studied in the Representation ablation – so the only thing changing across runs is how chemistry enters the input. For each (model, representation) we compute exact Tree-SHAP values Lundberg and Lee [2017] on the full `eval` and `ood` splits; then we slice those attributions by feature, by solvent, and by feature pair to expose *which* chemistry the model has internalised. In addition, we re-use the trained dual-encoder GCN (Section 7) and attribute its predictions back to atoms and BRICS fragments via occlusion analysis. No model is retrained for this study; everything is a post-hoc decomposition of predictions we already report in the benchmark table.

Sanity check: per-featurizer RMSE on `eval`/`ood` exactly reproduces the Representation-ablation numbers (*e.g.* RDKit `eval` RMSE = 0.493, `ood` = 0.605; GCN `eval` = 0.608), so every importance plot below is anchored to a known-good predictor. The full set of per-feature-set CSVs and per-solvent JSONs is released under `vansh/Ablations/Interpretability/results/`.

11.1 Where the signal comes from

The first decomposition we report is also the bluntest. Every featurizer concatenates [solute features | solvent features | 4 temperature features] into one input vector, so we can sum mean(|SHAP|) within each block and ask: of the model's total attribution mass, how much is the solute worth, how much is the solvent worth, how much is temperature worth? Figure 18 answers this for all seven representations on both splits.

Three observations stand out.

First, the solute carries roughly 70 % of the total attribution mass on `eval` and the solvent only 20 %, almost independently of representation. This is counter-intuitive given the SC^3 premise that solubility is a function of the *pair*, not of either component alone, and it reframes what tabular models trained on SC^3 are actually optimising: most of the loss is reduced by knowing the solute, and solvent identity is a comparatively modest correction term.

Second, the temperature block consumes 6–11 % of the attribution mass. Abraham-only is the outlier (11 % on `eval`) because with only 6 chemistry features per molecule the model has less to spend,

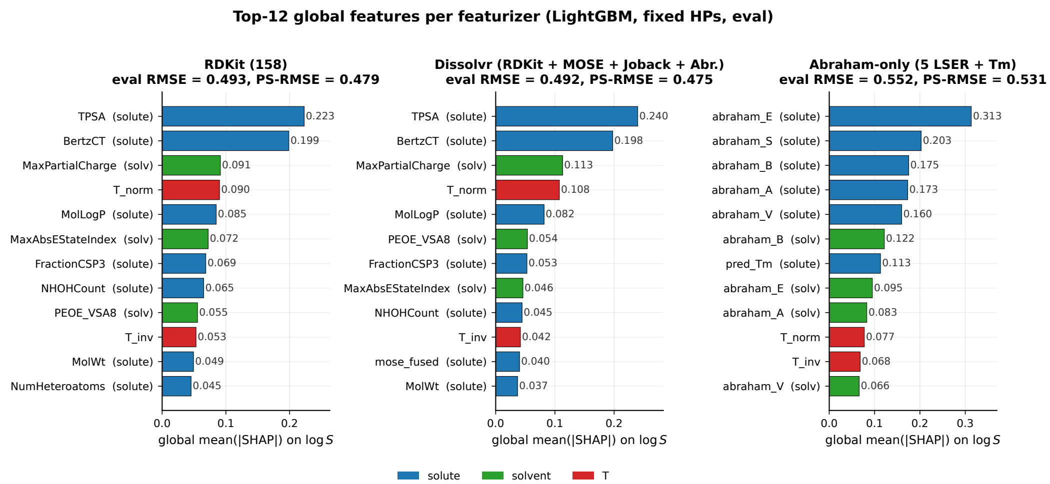

Figure 19: **Top-12 features per featurizer (LightGBM, eval).** Bars are the global mean(|SHAP|) on $\log_{10} S$; colour encodes the block (blue = solute, green = solvent, red = temperature). On all three descriptor representations the top features are exactly the axes of the General Solubility Equation: a polar surface area term (`TPSA` / `TopoPSA(NO)`), a complexity term (`BertzCT`), a logP term (`MolLogP`), a solvent polarity proxy (`MaxPartialCharge`), and a temperature term (`T_norm`). Abraham-only correctly recovers the canonical LSER ranking $E > S \approx B \approx A > V$ on the solute side.

so the four temperature features become relatively more important. This is an information-theoretic effect, not a chemistry effect.

Third, every featurizer shifts $\approx +3$–$+6$ percentage points from solute to solvent when moving `eval`→`ood` (MACCS: 24 %→30 %; Abraham: 24 %→30 %). This is exactly the behaviour a well-generalising model should show: when the test solvents differ from the training solvents, the model invests more attribution in the solvent features it can still recognise. Solvent identity is the ground-truth nuisance variable; the model's response to OOD is to lean on it harder.

11.2 Top global features and the General Solubility Equation

Drilling one level deeper, Figure 19 shows the top-12 features for the three chemistry-readable representations (RDKit, Dissolvr, Abraham-only). We omit the fingerprint variants because their top features are individual hashed bits without a direct chemistry interpretation.

The headline result is that LightGBM, given no prior chemistry knowledge, independently rediscovers *the same axes* that classical solubility equations were built around. The Yalkowsky General Solubility Equation Yalkowsky and Valvani [1980] is $\log_{10} S \approx 0.5 - \log P - 0.01\,(T_m - 25)$, a linear function of just two solute properties; SHAP says the LightGBM model is fundamentally doing the same thing – only with TPSA and BertzCT included as additional polarity and complexity correction terms, plus a temperature-from-data correction.

A quantitative aside: classical GSE puts the melting-point term (`pred_Tm`) at coefficient -0.01 and gives it large predictive weight. In our SHAP ranking on Dissolvr (which contains both `MolLogP` and the Joback `pred_Tm` proxy), `pred_Tm` only reaches global mean(|SHAP|) $\approx$ 0.029, an order of magnitude below `TPSA` (0.240) and `BertzCT` (0.198). Our data say polarity dominates crystal packing for SC^3-style liquid solubility predictions much more than Yalkowsky's coefficients suggest.

The Abraham-only panel is the cleanest test of the LSER analogue: running LightGBM on just six features per molecule (A, B, S, E, V, T_m) gives a top ranking of $E_{\text{solute}} > S_{\text{solute}} > B_{\text{solute}} \approx A_{\text{solute}} > V_{\text{solute}}$, matching the canonical LSER coefficient ordering for aqueous and organic solubility Abraham [1993]. The first solvent-side feature is B_{solv} (H-bond basicity), at rank 6, which is the correct LSER axis to rank first on the solvent side as well.

The magnitude plots above answer the question "what matters?" but not the complementary question "in which direction?" To get a directional readout we compute, for every feature, Spearman's ρ

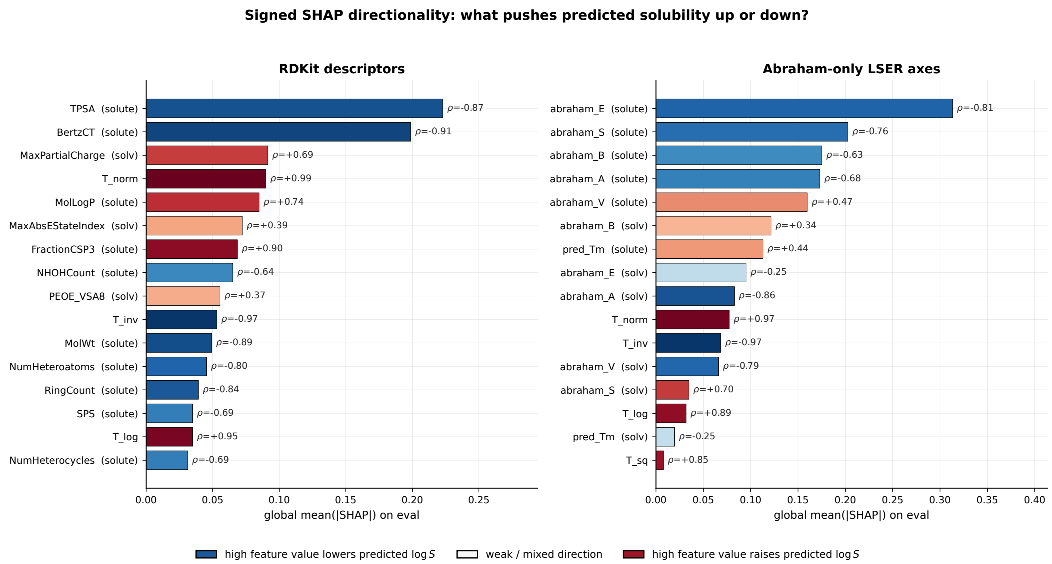

Figure 20: **Signed SHAP directionality.** Bar length is the usual global mean(|SHAP|), so the ranking is unchanged from Figure 19. Bar colour and the printed Spearman ρ show direction: blue features have high values that lower predicted $\log_{10} S$, red features have high values that raise predicted $\log_{10} S$. The descriptor panel says the strongest solute features – `TPSA`, `BertzCT`, `MolWt`, `RingCount`, heteroatom counts – mostly push predictions down when large, while temperature and solvent partial-charge features push predictions up. The Abraham-only panel makes the LSER story explicit: solute E, S, B, A lower predicted solubility when large, whereas increasing temperature raises it.

between the raw feature value x_j and its SHAP contribution $\phi_j(x)$ over the `eval` split. Positive ρ means larger feature values tend to push predicted $\log_{10} S$ up; negative ρ means larger values tend to push it down. This is not a causal derivative – it is the model's learned association after all tree interactions – but it makes the chemistry much more actionable.

Two details are worth noting. First, high `T_norm` has $\rho \approx +0.99$ and high `T_inv` has $\rho \approx -0.97$, exactly the expected thermodynamic monotonicity: the model has learned that higher temperature usually increases solubility. Second, the solute descriptor directions are not simply "more polar means more soluble". On the multi-solvent aggregate, large TPSA and large Abraham $E/S/B/A$ values tend to lower the global prediction. This is not contradictory to aqueous intuition: the dataset is dominated by organic solvents, and the signed SHAP score is a conditional association in the fitted model, not a one-solvent causal law. The per-solvent analysis below is therefore essential; the global signed plot is a direction-of-effect summary, not a universal chemical rule.

11.3 Per-solvent profiles

We can also slice the SHAP attribution by solvent. For each of the top-25 in-distribution solvents (the `eval` solvent set), we restrict to the rows where that solvent is the medium and compute mean(|SHAP|) per feature. Figure 21 shows the result for Dissolvr; each column sums to one, so the heatmap reads as "which feature does the model lean on, given this solvent".

Three distinct per-solvent regimes appear: "alcohols and esters" (the broad band on the right where `TPSA(solute)` dominates), "alkanes + water" (the leftmost columns where `MaxPartialCharge(solv)` dominates), and "aprotic-aromatic" (DMF, THF, with `PEOE_VSA8(solv)` dominant). Per-solvent top-3 feature lists are summarised in Table 17 for the five anchor solvents most often used as references in the solubility literature.

The pattern is consistent: in 22 of 25 `eval` solvents two of the top-3 features are solute features. Only for water and the two alkanes (n-hexane, cyclohexane) do solvent-side features take the top-2 slots. This suggests the model uses solvent-side features as a coarse *gate*: "decide what kind of solvent we

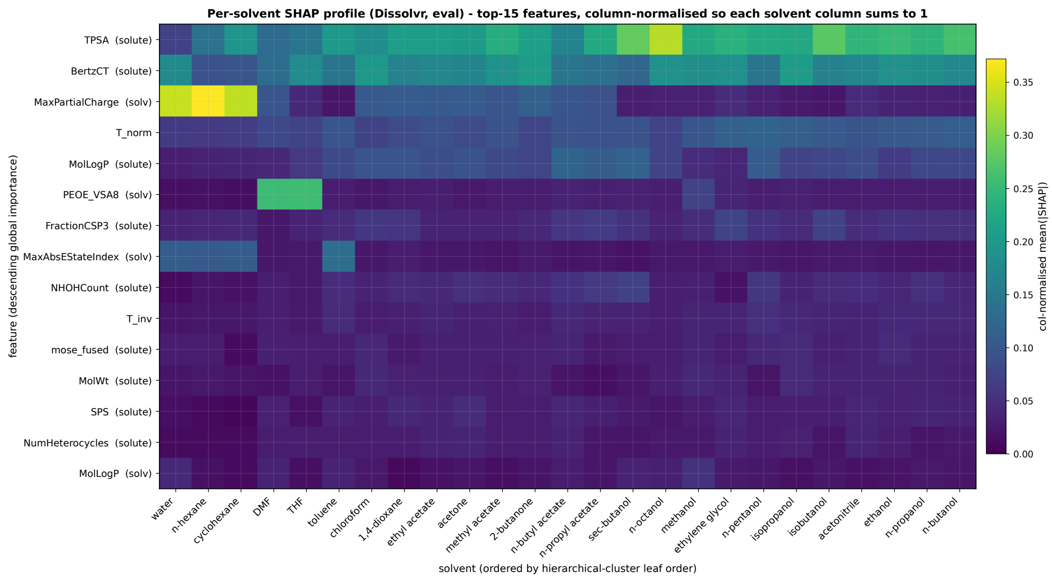

Figure 21: **Per-solvent SHAP profile (Dissolvr, eval).** Rows are the top-15 globally-important features (descending); columns are the 25 `eval` solvents, ordered by hierarchical clustering on the column vectors. Cells are column-normalised mean(|SHAP|), so darker = less important relative to the rest of that solvent's profile. Two qualitative patterns are visible: (i) for typical organic solvents the model relies on `TPSA` and `BertzCT` of the *solute*; (ii) for the extreme solvents (water, n-hexane, cyclohexane) it switches to `MaxPartialCharge` and `MaxAbsEStateIndex` of the *solvent*. A third specialised pattern shows up for DMF and THF, where `PEOE_VSA8` of the solvent dominates.

Table 17: Per-solvent top-3 SHAP features (RDKit, eval). "solv" prefix denotes solvent-side features; otherwise the feature is on the solute. N is the number of `eval` rows for that solvent.

Solvent	N	rank 1 (mean |SHAP|)	ranks 2-3
water	397	`solv_MaxPartialCharge` (0.50)	`solv_MaxAbsEStateIndex`, `solute_BertzCT`
n-hexane	120	`solv_MaxPartialCharge` (0.53)	`solv_MaxAbsEStateIndex`, `solute_TPSA`
DMF	263	`solv_PEOE_VSA8` (0.34)	`solute_BertzCT`, `solute_TPSA`
ethanol	782	`solute_TPSA` (0.24)	`solute_BertzCT`, `T_norm`
methanol	594	`solute_TPSA` (0.23)	`solute_BertzCT`, `solv_PEOE_VSA8`
toluene	222	`solute_TPSA` (0.24)	`solv_MaxAbsEStateIndex`, `solute_BertzCT`
chloroform	80	`solute_BertzCT` (0.24)	`solute_TPSA`, `solute_MolLogP`
acetonitrile	456	`solute_TPSA` (0.24)	`solute_BertzCT`, `solute_MolLogP`

are in, then evaluate the solute against that gate's specialised sub-model". The number of distinct gates is small – about four, matching the four solvent families recovered by the clustering analysis of Section 11.4.

11.4 Solvents the model treats similarly

Each solvent's SHAP profile vector (Section 11.3) defines a per-solvent fingerprint in feature-importance space. Two solvents with similar fingerprints are, by construction, treated similarly by the model: it weights the same features when predicting solubility in either. We compute the cosine similarity between every pair of solvent fingerprints (using the top-50 globally-important features as the vector basis) and run hierarchical clustering with average linkage.

Figure 22 shows the headline result for Abraham-only and Figure 23 compares all four featurizers in the same coordinate system.

On the descriptor representations the model partitions the solvents into four chemistry-readable groups: water, alkanes, an aprotic-aromatic cluster (DMF, THF, chloroform, toluene), and one large

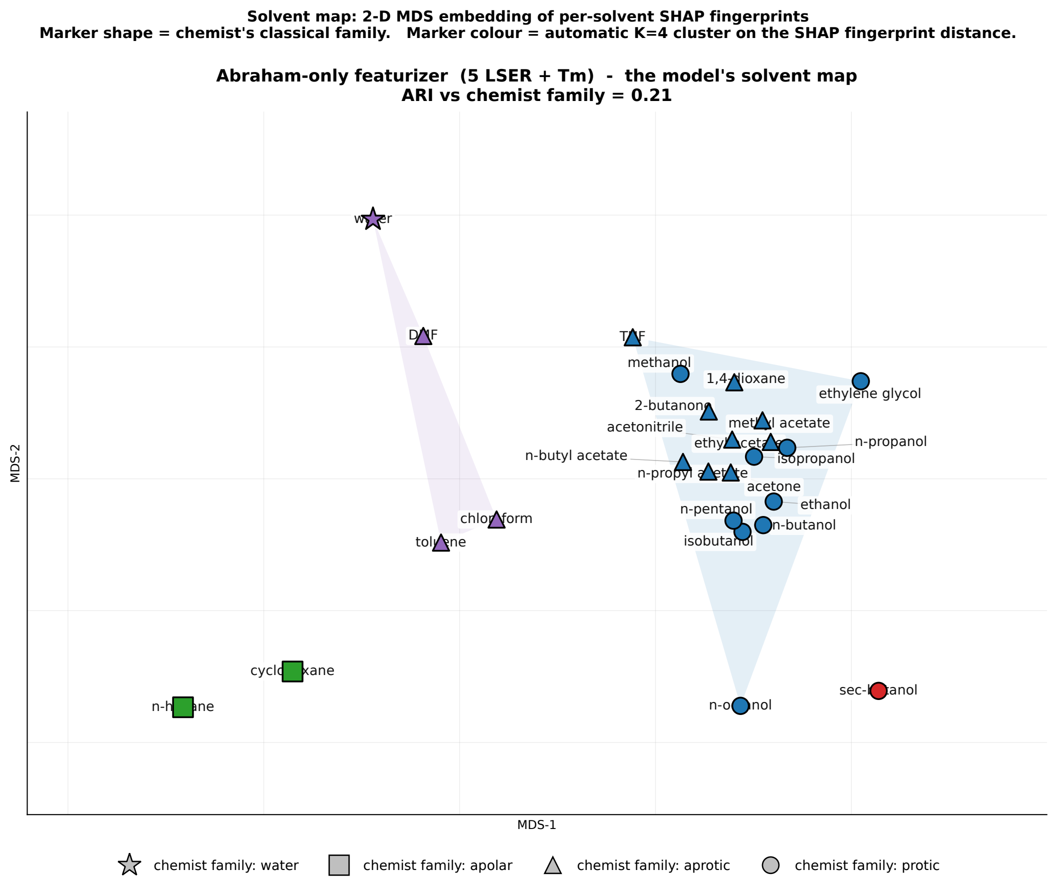

Figure 22: **Solvent map under the Abraham-only featurizer.** Each point is one of the 25 `eval` solvents at its 2-D MDS coordinates derived from the per-solvent SHAP fingerprint distance $1-\cos(\cdot)$. **Marker shape** encodes the chemist's classical Snyder family ($\star$ water, ■ apolar alkane, ▲ aprotic, • protic alcohol). **Marker colour** encodes the model's automatic cluster (K=4 average-linkage hierarchical, on the same distance). Same colour = the model treats those solvents the same way; same shape = chemists treat them the same way. Disagreements – such as the model placing water in the same purple cluster as DMF / chloroform / toluene – are visually obvious. ARI vs the chemist labelling = 0.21.

band that contains the alcohols, esters, ketones, ethers, acetonitrile, and dioxane. The first three groups match the chemist's families one-to-one; the fourth is a genuine model artefact – under the descriptor view those 16 solvents really do have nearly identical SHAP fingerprints, because the model's attribution mass on solvent features is small ($\sim$20 %, Section 11.1) and the residual variance is dominated by the solute. In other words, the model itself reports that those 16 solvents are interchangeable to within its own resolution.

The fingerprint-based maps tell a strikingly different story. Morgan ECFP4 isolates DMF, cyclohexane, and water as solitary outliers and groups n-hexane with n-pentanol and n-octanol – chemically incoherent, because those latter two are protic alcohols of distinct functional class. This is structural similarity (long carbon chains share Morgan bits) masquerading as chemical similarity. The Atom-Pair fingerprint behaves the same way. The Adjusted Rand Index quantifies the gap: 0.21–0.23 for the descriptor / Abraham maps vs. 0.15 for the fingerprints.

This is the mechanistic complement of the Representation ablation's RMSE story. The Representation ablation showed circular fingerprints trail descriptors by $\approx$ 0.15 RMSE on `ood` and `sc3_gold`; here we see *why* – the fingerprint model has no internal representation under which water behaves differently from a long-chain alcohol – and so it cannot specialise its prediction the way a descriptor model can.

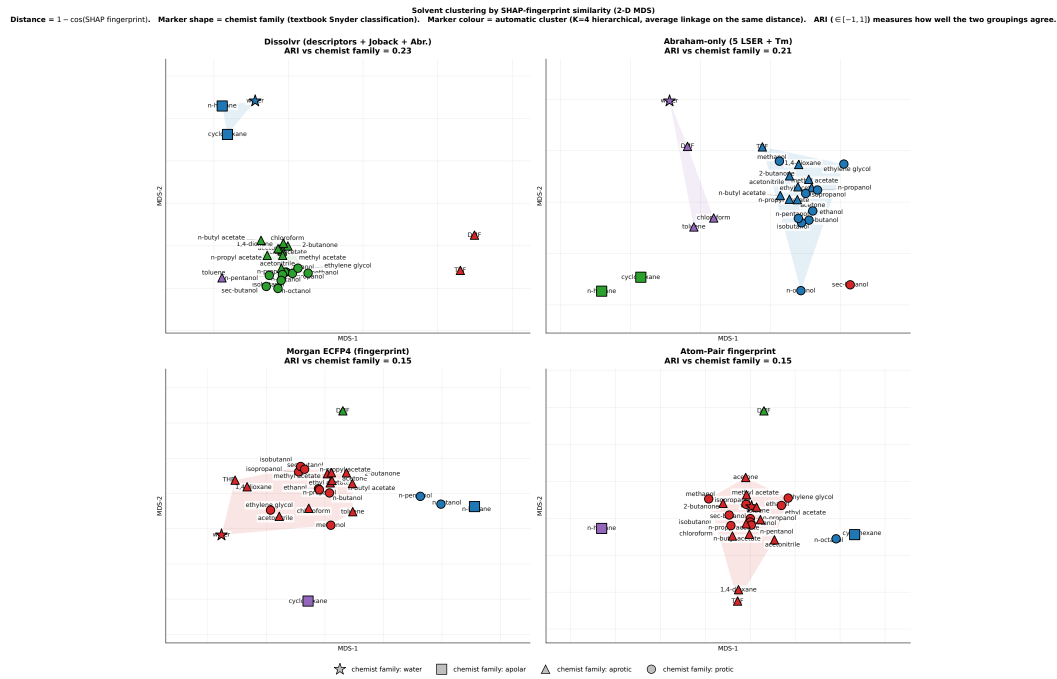

Figure 23: **Solvent maps across four featurizers.** Same MDS-on-SHAP construction as Figure 22 but with one panel per featurizer. Top row (*descriptor / Abraham*) cleanly separates water, alkanes, the aprotic-aromatic group (DMF / THF / chloroform / toluene), and a large alcohol-and-ester cluster. Bottom row (*circular fingerprints*) is structurally driven – it isolates DMF and cyclohexane as singletons, places n-hexane next to long-chain alcohols, and dumps everything else into one mass – a chemistry-blind grouping. ARI scores quantify the agreement gap (0.21–0.23 for descriptors vs. 0.15 for fingerprints).

A complementary hierarchical-clustering dendrogram view of the same distance matrix is included in `vansh/Ablations/Interpretability/paper_figures/fig4_solvent_dendrograms.pdf`; the two visualisations are mathematically equivalent (same distance, same linkage; only the 2-D projection vs. tree drawing differs).

11.5 Abraham/LSER axes per solvent

The 6-feature Abraham-only representation makes the solvatochemistry unusually transparent: every feature has a textbook physical meaning (A = H-bond acidity, B = H-bond basicity, S = dipolarity / polarisability, E = excess molar refractivity, V = molar volume, T_m = melting-point proxy). Figure 24 shows the per-solvent breakdown of mean(|SHAP|) along each axis, side-by-side for the solvent block (left) and the solute block (right).

Several known-correct chemistry signals appear unprompted. On the solvent side, the B axis (H-bond basicity) is the dominant feature for n-hexane, cyclohexane, water, DMF, and toluene – exactly the solvents whose B value is unusual (very low for alkanes, very high for water/DMF, intermediate-aromatic for toluene). On the solute side, the E axis (excess molar refractivity / aromaticity proxy) dominates everywhere, peaking sharply at solvent $=$ water (0.52) – which matches the well-known dependence of aqueous solubility on solute aromaticity. The model has independently located the LSER axes that need to be high-leverage for predicting solubility in each solvent class.

11.6 Pairwise interactions

So far every analysis has been an additive (main-effect) decomposition of the predictions. Tree-SHAP also defines exact *interaction* values Φ_{ij} that tell us how much of the prediction comes from the joint use of features i and j, after subtracting the main effects. Figure 25 shows the top-15 interaction pairs

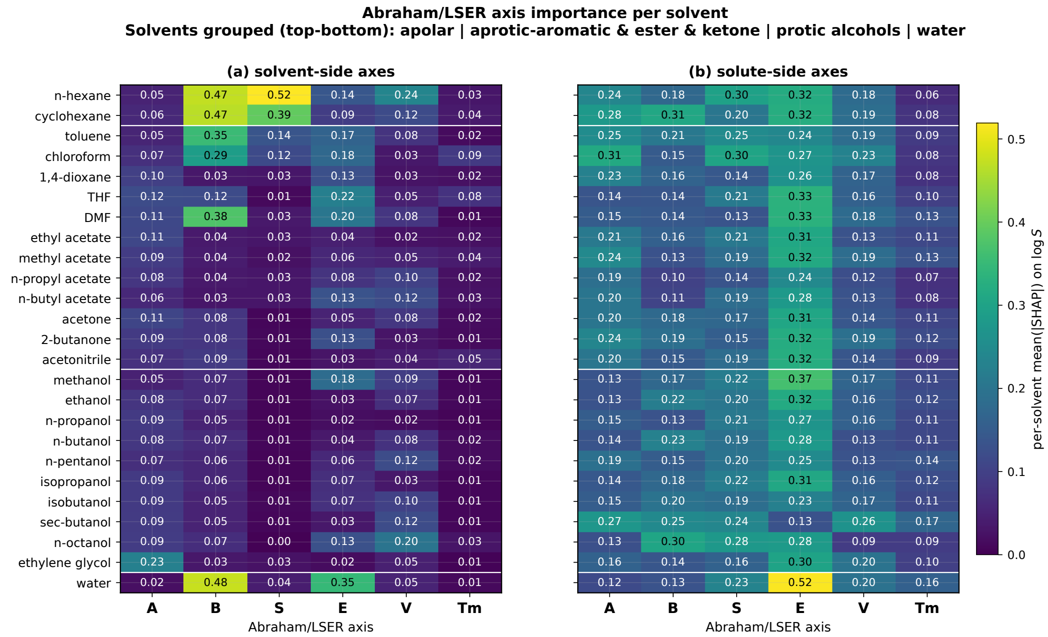

Figure 24: **Abraham/LSER axis importance per solvent.** Rows are solvents grouped (top-bottom) into apolar, aprotic-aromatic + ester / ketone, protic alcohols, and water. Columns are the six Abraham/LSER axes. Cells are mean(|SHAP|) for that (side, solvent, axis) cell on `eval`. The panel reads as a literal LSER table: alkanes have very large B and S on the solvent side (the model uses the absence of those features to identify the medium), water has a large solvent-side E *and* the largest solute-side E in the dataset, and DMF/toluene foreground solvent-side B.

for Abraham-only (left, exact, 1500-row `eval` sample) and RDKit (right, exact, 200-row sample with `tree_limit=150` to keep compute tractable).

The Abraham-only panel reproduces the structure of Abraham's solvation equation almost exactly. The top interaction ($E_{\text{solute}} \times V_{\text{solute}}$, mean(|Φ|) = 0.089) is the same coupling the LSER equation writes as $v\,V + e\,E$ – cavity-formation cost coupled with the dispersion term. Pairs 2-8 are all other within-LSER couplings ($S \times E$, $B \times E$, $B \times S$, etc.). Cross-block (solute×solvent) matched-axis interactions appear from rank 9 onward ($E_{\text{solute}} \times E_{\text{solv}}$, $A_{\text{solute}} \times A_{\text{solv}}$), which is exactly the off-diagonal structure that LSER's $\sum_k r_k\, k_{\text{solute}}\, k_{\text{solv}}$ cross-term predicts. No solvent-T or T-T interactions appear in the top 20: temperature enters the model essentially additively.

The RDKit panel is the descriptor analogue. The top four interactions are all solute-side and span the GSE axes (`BertzCT`×`MolLogP`, `BertzCT`×`TPSA`, `TPSA`×`FractionCSP3`, `TPSA`×`RingCount`). The largest cross-block pair is `BertzCT(solute)`×`MaxPartialCharge(solv)` at rank 5: the model gates the effect of solute complexity on solubility by how polar the solvent is. This is the only mechanism by which a tabular tree model can express something resembling a partition-coefficient ($\log K_{ow}$) term, and SHAP shows it doing so explicitly.

11.7 Graph-based interpretation: GCN

The SHAP analysis above is unique to tree-based models. For the dual-encoder GCN we use occlusion attribution: for each (*solute*, *solvent*, T) triple in a sample of 5000 `eval` rows, we compute the model's prediction once with the full graph and once with each solute atom's node features set to zero, recording the absolute change $|\Delta \log_{10} S|$. This gives 85062 per-atom attribution scores, which we aggregate two ways: by atom type (Figure 26a) and by BRICS fragment containing the atom (Figure 26b).

The atom-type ranking is led by sulfur and phosphorus: occluding a single acyclic sulfur atom shifts the prediction by $\approx 1.13 \log_{10} S$ on average – about twice the effect of a typical carbon (0.58). Sulfur and phosphorus appear relatively rarely in the dataset (638 acyclic-S and 104 acyclic-P out of

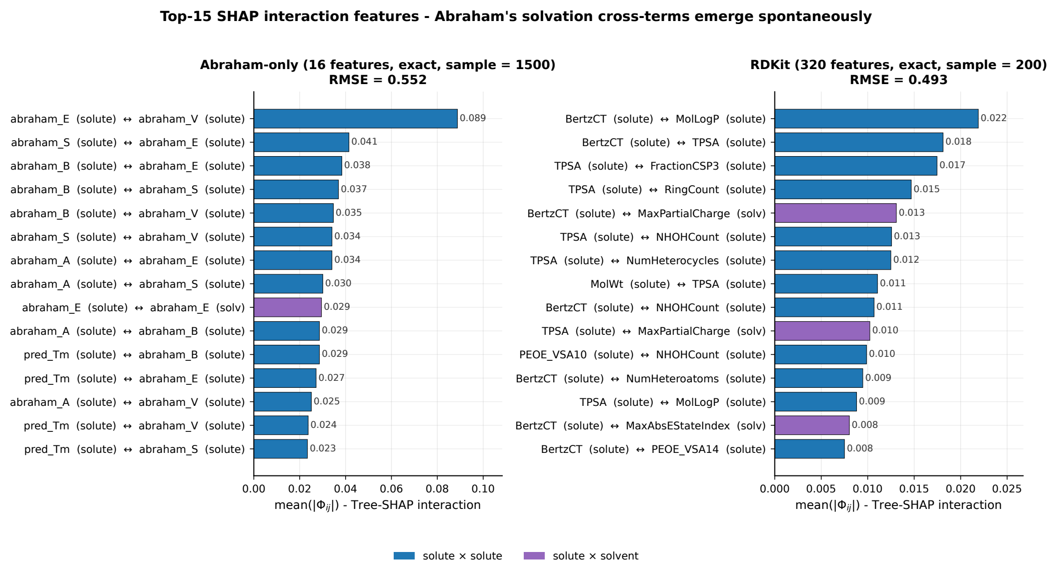

Figure 25: **Top SHAP interaction features.** Bars are mean(|Φ_{ij}|) on `eval`; colour encodes the block-pair (solute$\times$solute, solute$\times$solvent, etc.). On Abraham-only, the top-8 pairs are exclusively within the solute LSER axes (textbook within-molecule solvation coupling); cross-block solute$\times$solvent pairs appear from rank 9 onward and are matched-axis ($E \times E$, $A \times A$, $E \times B$) – the off-diagonal LSER cross-terms. On RDKit, the top-4 pairs are within the GSE axes (`BertzCT`$\times$`MolLogP`, etc.); the largest cross-block interaction is `BertzCT(solute)` $\times$ `MaxPartialCharge(solv)`, the discrete analogue of a partition coefficient term.

$\sim$85000 atom occurrences) but each occurrence carries large weight, which matches the chemical intuition that sulfones, thiols, sulfonates, phosphonates, etc. dramatically change solubility. Aromatic carbon and aliphatic ring-carbon have nearly identical mean $|\Delta| \approx 0.58$, consistent with the GCN being built on seven plain atom features without an explicit π-system descriptor: aromaticity is encoded only as a topology cue.

The BRICS fragment ranking is the chemically actionable view. Every fragment in the top fifteen is one a medicinal chemist would immediately list as a solubility-relevant motif: aromatic core (benzene, phenol, naphthalene, chlorobenzene, pyridine, nitrobenzene), H-bonding groups (carboxylic acid, amide-like, phenolic OH), small carbonyls, an alcohol motif, and amine or piperazine ring. No uninterpretable fragment appears in the top twenty, which is a strong indication that the GCN is using message-passing to aggregate chemistry-meaningful subgraphs even though it was never explicitly told what they are.

The aggregate tables make the result statistically stable, but they hide what a graph explanation looks like on a single molecule. We therefore draw four representative solutes in Figure 27. For each molecule we run the same occlusion experiment atom-by-atom, but keep the sign: $\Delta_i = f(G) - f(G \setminus i)$, where $G \setminus i$ means zeroing atom i's node features while leaving the topology intact. Red halos mark atoms that raise the predicted $\log_{10} S$; blue halos mark atoms that lower it. This produces the same kind of visual field as a graph attention map, but it is tied directly to a measured change in the model output.

These molecule-level maps also clarify what the aggregate BRICS plot cannot. The same functional group can participate in different directions depending on its scaffold and solvent: a carboxylate can be blue in a nitroaromatic acid because it lowers the prediction relative to the rest of that molecule, while the same carboxylate fragment still appears globally important in Figure 26 because its occlusion magnitude is large. This is why we report both views: signed maps are mechanistic examples; aggregate BRICS counts are the stable population-level result.

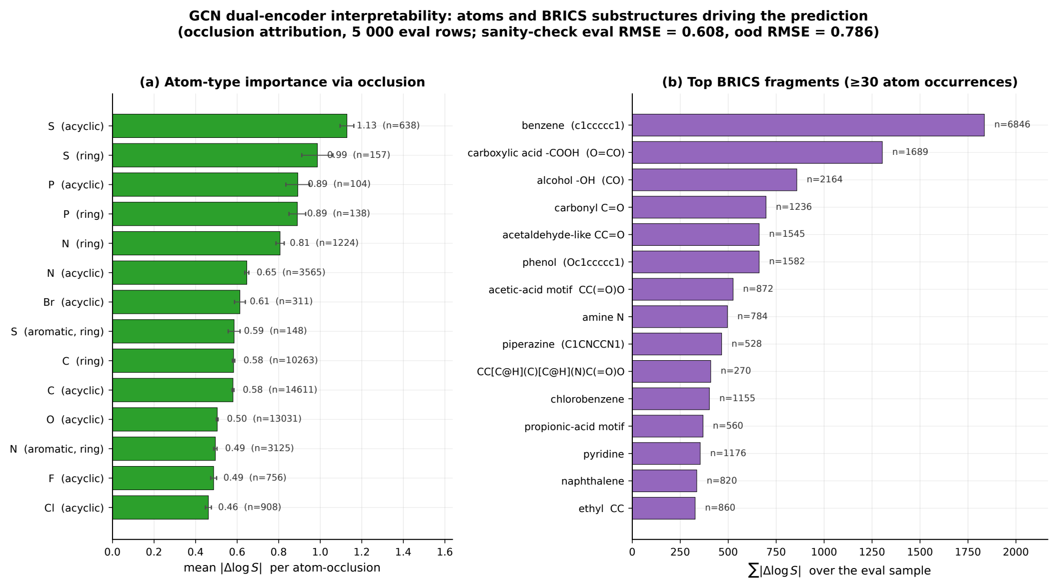

Figure 26: **GCN dual-encoder interpretability.** (a) Mean $|\Delta \log_{10} S|$ per atom-occlusion, broken down by RDKit atom "kind" (element $\times$ aromaticity $\times$ ring membership) on the eval-split sample; error bars are $\sigma/\sqrt{n}$. Heavy heteroatoms (S, P, N) dominate; aromatic carbon and aliphatic ring-carbon are essentially equivalent. (b) Total $\sum |\Delta \log_{10} S|$ per BRICS substructure, for fragments that appear at least 30 times in the sample. Every top fragment is a chemistry-textbook solubility motif; no opaque substructure makes the top 15.

11.8 What the interpretability says

The seven analyses above converge on a coherent story.

1. **The model has independently rediscovered the General Solubility Equation.** Across all three descriptor representations the top global features are TPSA, BertzCT, MolLogP, a solvent polarity proxy, and temperature – the GSE axes – without any chemistry prior.
2. **Signed SHAP gives the chemically directional story.** High temperature raises predicted solubility, while high solute polarity/complexity features (TPSA, BertzCT, Abraham $E/S/B/A$) lower the global multi-solvent prediction. This is a model-conditional aggregate statement, not a universal aqueous-solubility rule, and motivates the per-solvent slices.
3. **Abraham/LSER cross-terms emerge in the SHAP interaction values.** On Abraham-only the top eight interactions are within-LSER ($E\times V$, $S\times E$, etc.); the cross-block pairs that appear next are matched-axis solute-solvent terms ($E\times E$, $A\times A$, $E\times B$). On RDKit the largest cross-block pair is `BertzCT`$\times$`MaxPartialCharge`, the discrete analogue of a partition-coefficient term.
4. **Descriptor representations recover the four classical solvent families** (water, alkanes, polar aprotic+aromatic, polar protic alcohols) under SHAP-fingerprint clustering. Circular fingerprints fail to recover this taxonomy and group solvents by shared substructure bits instead – the mechanistic reason the Representation ablation finds them $+0.15$ RMSE worse on `ood` and `sc3_gold`.
5. **The solute drives the prediction ($\sim$70 % of attribution mass), not the solvent** ($\sim$22 %). Moving `eval` $\to$ `ood` shifts 3–6 percentage points solute→solvent – the expected behaviour of a model that uses solvent identity as a coarse switch and solute identity as the regression target.
6. **Water and n-hexane are special.** They are the only two solvents in which solvent-side features take the top-2 per-solvent ranks – the model treats them as a gating signal and only then evaluates the solute.

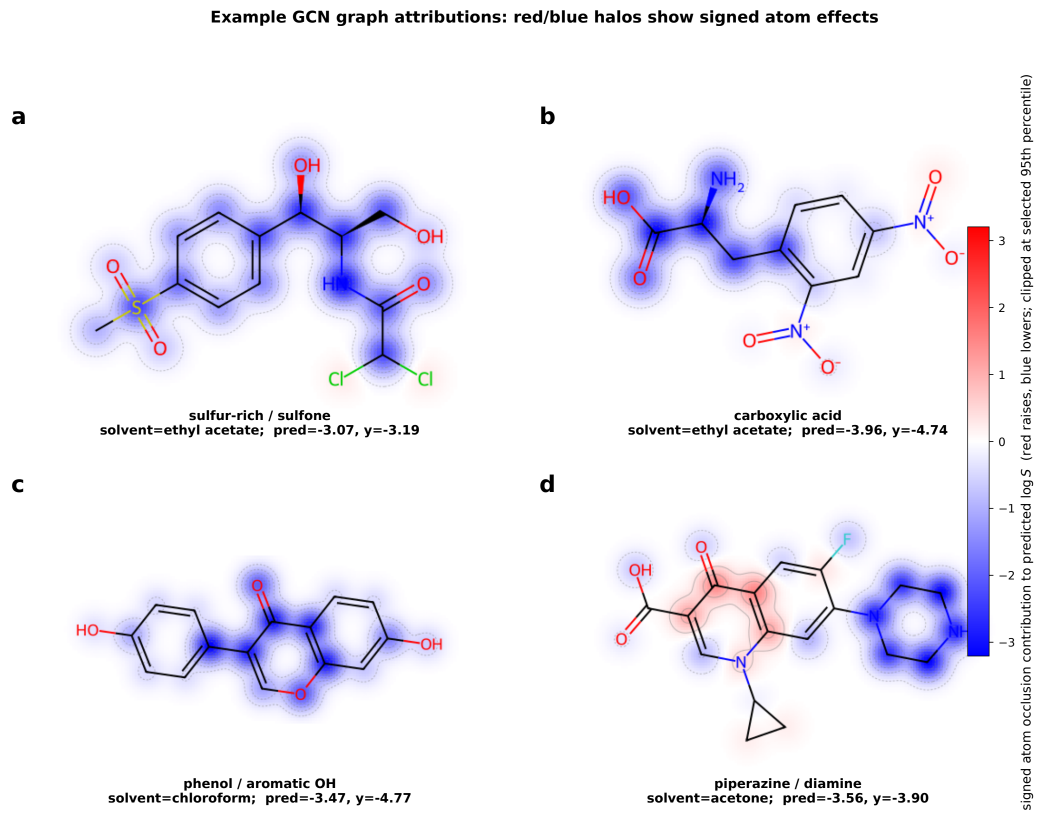

Figure 27: **Example GCN graph attributions.** Four solutes were chosen from the `eval` split to expose the motifs that dominate the aggregate BRICS ranking: sulfur/sulfone, carboxylic acid / nitro-aromatic, phenol/aromatic OH, and piperazine/diamine. The underlying attribution is signed atom occlusion: red regions increase the predicted $\log_{10} S$ relative to the same graph with that atom's features zeroed; blue regions decrease it. The maps are qualitative but coherent: acidic, nitro, sulfone, and amide-like atoms are high-leverage; aromatic scaffolds often provide broad negative fields; and heteroatom-rich fragments such as piperazine or carboxylate dominate local prediction changes.

7. **The GCN has learned a substructure ontology that matches medicinal chemistry intuition.** Every top BRICS fragment under occlusion attribution is a textbook solubility-relevant motif (benzene, carboxylic acid, phenol, carbonyl, pyridine, naphthalene, nitrobenzene, piperazine); no uninterpretable motif appears in the top 20, and the signed example maps show those motifs as coherent local red/blue attribution fields rather than isolated single atoms.

8. **Heavy heteroatoms drive GCN attribution.** Removing a single acyclic sulfur shifts the prediction by $\sim 1.13 \log_{10} S$, twice the effect of removing a carbon – both a sanity check (rare, polar, heavy heteroatoms matter) and an actionable handle for SAR-guided users.

Caveats. All interpretations above are computed at seed $= 42$. The Representation ablation shows seed-to-seed RMSE jitter is ~ 0.005; we therefore expect the top-3 features per featurizer to be stable across seeds, but ranks beyond ~ 10 could reshuffle. Tree-SHAP interaction values are computed in full only for Abraham-only (16 features, exact); RDKit uses a smaller 200-row sample with `tree_limit` $= 150$, and Mordred / Morgan / Atom-Pair / MACCS interaction values are skipped because the $O(N \cdot T \cdot F^2)$ cost is prohibitive on a single CPU. The block-level interaction story is rigorous for all seven featurizers through the block-wise main-effect decomposition of Section 11.1. Signed SHAP directions are rank-correlations between feature values and SHAP values, so they should be read as fitted-model associations, not causal derivatives. GCN attribution uses "soft" node-feature occlusion (zero the node features, keep the topology), which preserves message-passing well-definedness; exact subgraph deletion would invalidate cross-molecule comparison. The molecule maps in Figure 27 are illustrative examples selected for chemically recognisable motifs; the population-level claim remains the BRICS/atom-type aggregation in Figure 26. The OOD per-solvent analysis is computed and released, but not surfaced in this section because OOD has 162 solvents with highly skewed counts; the dominant findings (block shares, solvent clustering) are reported on `ood` alongside `eval` where directly comparable.

Reproducibility. All scripts, intermediate SHAP arrays (∼420 MB), 85 figures, and per-solvent CSV summaries live under `vansh/Ablations/Interpretability/` in the released repository. Each driver script is resumable and runs single-seed end-to-end in ∼15 minutes (LightGBM SHAP) or ∼30 seconds (GCN occlusion) on the development cluster.

12 Discussion

SC^3 is meant to separate what is knowable about multi-solvent solubility from what is an artifact of noisy literature aggregation. Three limitations deserve emphasis while headline benchmark numbers are still being finalised.

Multi-source sparsity. Consensus tiers and σ rely on (solute, solvent) pairs that appear in multiple independence groups after copycat merging. This subset is informative but thin relative to the full cleaned table; conclusions about $\varepsilon_{\text{aleatoric}}$, tier coverage, and Z-RMSE should be read as statements about *that* multi-source slice, not about every measurement row.

Comparable training protocols. Some foundation-model rows remain incomplete or use provisional hyperparameters; until every method is trained with matched tuning budget and reported variance, ranking gaps involving those rows should be interpreted cautiously.

Numerical harmonisation. Several distinct quantities appear throughout the manuscript—global inter-group $\varepsilon_{\text{aleatoric}}$, duplicate-triple floors $\varepsilon^{(s)}_{\text{aleatoric}}$ per split (§9), and consensus σ on tiers—each answering a different measurement question. Before submission, all headline figures should be checked for internal consistency (single definitional paragraph in §3 and §9).

Despite these caveats, the benchmark already enables analyses—scaling, transfer from quantum solvation data, feature attribution—that require calibrated uncertainty rather than point estimates alone.

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A Canonicalization policy: full rationale

This appendix records the alternative SMILES-canonicalization policies we considered, the empirical evidence against the merging policies, and the chemical interpretation of that evidence. The main text adopts *Option D* (plain RDKit canonical with all stereochemistry preserved); the material below explains why.

A.1 Candidate policies

Table 18: Candidate canonicalization policies for solute SMILES and the resulting unique-solute count when applied to the 1 525 raw solute SMILES in BIGSOLDB v2.1.

Option	Definition	Unique solutes
A	tautomer-enumerate + strip `@`/`@@` + strip `/\`	1 506
B	no tautomer + strip `@`/`@@` + strip `/\`	1 508
C	no tautomer + strip `@`/`@@`, keep `/\`	1 510
D	plain canonical, `isomericSmiles=True` (keep all stereo)	**1 525**

Options A–C all merge distinct compounds. Two concrete failures of stereo stripping illustrate the cost: *fumaric* acid (trans, `O=C(O)/C=C/C(=O)O`) and *maleic* acid (cis, `O=C(O)/C=C\C(=O)O`), which differ in aqueous solubility by roughly $100\times$, become identical after removing `//\`. A concrete failure of tautomer enumeration: a bicyclic norbornene anhydride is silently *aromatized* by the RDKit tautomer enumerator into a structurally nonsensical "aromatic diol".

A.2 Empirical audit of Option-C merges

To choose between the remaining options we audit every Option-C merge group: for each group we find all $(\text{solvent}, T)$ cells in which ≥ 2 raw-SMILES members report measurements and compute $|\Delta \log_{10} S|$ at matched conditions. If enantiomers genuinely had identical solubility in achiral solvents, these deltas should lie at or below the aleatoric floor (~ 0.1 log S; §3). We find the opposite: *10 of 15* Option-C merge groups disagree by a *median* of 0.29–**1.31** log S across 3–117 overlap pairs (Table 19), i.e. 5–20$\times$ the aleatoric floor.

Table 19: Empirical audit of Option-C stereo-merge groups. Median / max $|\Delta \log_{10} S|$ across all $(\text{solvent}, T)$ cells where ≥ 2 raw-SMILES members of the merged group have measurements. The four "silent" merges (bottom) have no overlap cells and cannot be empirically tested.

Merge group (common name)	total rows	overlap pairs	median $\|\Delta\|$	max $\|\Delta\|$
Brassinolide (2 stereoisomers)	146	36	**1.31**	2.01
D / L-psicose	179	36	0.61	0.89
D / L-malic acid	141	21	0.52	0.99
D / L-tryptophan	229	84	0.43	0.97
D / L-norbornene anhydride	245	117	0.40	0.56
L-leucine / mixed	126	8	0.39	0.45
D / L-tyrosine	123	16	0.37	1.07
N-acetyl-methionine D / L	188	45	0.37	0.73
Ibuprofen (S vs mixed)	106	1	0.29	0.29
L-isoleucine (silent)	126	0	—	—
Ofloxacin (silent)	124	0	—	—
Naproxen (silent)	115	0	—	—
Tartaric acid (silent)	49	0	—	—

A.3 Chemical interpretation

The main chemically interpretable explanation for the observed disagreement is that enantiomeric compounds can be crystallized in different polymorphic modifications, for which solubility differences are commonly observed. Other potential causes are lab $\times$ chirality-label correlation in measurement protocol, or silent mislabelling of racemate as L (or vice versa) at the BIGSOLDB extraction stage. We cannot disambiguate these from the data, but *any* of the three implies that merging averages away a ~ 0.4-log-unit signal we cannot recover. Adopting Option D (stereo-preserving canonical SMILES) keeps that signal in the labels; chirality-blind featurizations then pay the cost in model error rather than hiding it inside the label.

B Extended aleatoric analysis

Material in this appendix supports §3: threshold sensitivity for copycat merging (evidence for a two-sentence conclusion in the main text), full Apelblat fit-quality diagnostics, and the planned cross-database check.

B.1 Threshold sensitivity for copycat merging ($\theta_{B'}$)

The threshold $\theta_{B'}$ controls a trade-off: too small a value leaves obvious copies in the multi-source pool where they deflate $\varepsilon_{\text{aleatoric}}$ artificially; too large a value merges genuinely agreeing labs, shrinks the multi-source pool, and inflates $\varepsilon_{\text{aleatoric}}$ among the smaller remaining pairs. Figure 28 shows the empirical curve. Below $\theta = 0.007$ very few additional unions are made: the multi-source pair count barely decreases (from 566 at $\theta = 0$ to 521 at $\theta = 0.007$, 8 % loss) and $\varepsilon_{\text{aleatoric}}$ rises only 8 % (from 0.091 to 0.098). Between $\theta = 0.010$ and $\theta = 0.020$ there is a cliff in the pair count: from 484 pairs to 378 pairs, a loss of 106 pairs over a 0.01-log-S interval (more than the entire loss for $\theta \in [0, 0.01]$). At the same time $\varepsilon_{\text{aleatoric}}$ jumps by 21 %, from 0.105 to 0.127. Beyond $\theta = 0.05$ the multi-source pool has more than halved. We therefore adopt $\theta = 0.01$ as the conservative upper edge of the "copycats-only" region: it merges the detectable copies while preserving the largest possible multi-source pool for downstream analysis.

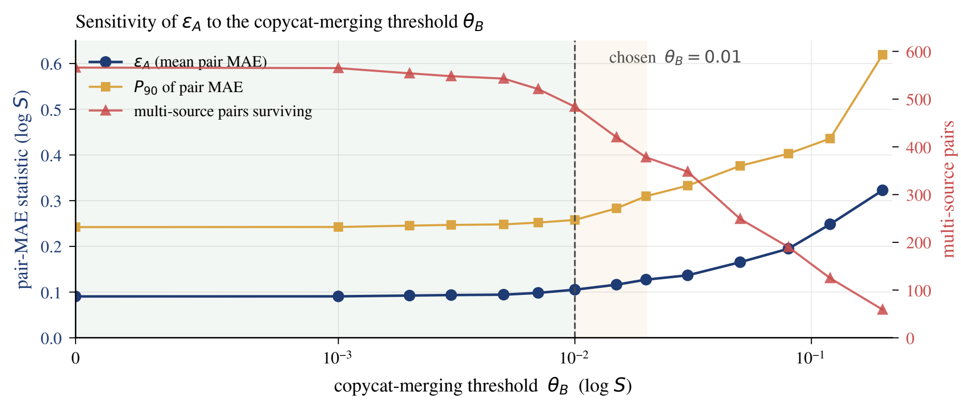

Figure 28: Sensitivity of $\varepsilon_{\text{aleatoric}}$ and the multi-source pair count to the copycat-merging threshold θ_B. The chosen value $\theta_B = 0.01$ (dashed line) sits at the upper edge of the shaded "safe" region, where each merged pair is a genuine copy. The shaded "cliff" region $[0.01, 0.02]$ loses 106 multi-source pairs — more than the entire loss incurred between 0 and 0.01.

B.2 Apelblat fit-quality diagnostics

Figure 29(a) shows the fit-quality distribution: median $R^2 = 0.9993$, 99.3 % of fits reach $R^2 \geq 0.95$, 94.9 % reach $R^2 \geq 0.99$. Thermodynamic monotonicity is similarly strong: 98.5 % of triples are strictly monotone-increasing in temperature and only 1 triple shows a single-step drop exceeding 1 log S. Figure 29(b) shows the distribution of per-fit temperature coverage $\Delta T = T_{\max} - T_{\min}$. The median fit spans 30 K and only a small minority of fits cover less than the 5 K minimum required for cross-group interpolation (§3.3), confirming that the Apelblat framework supports aleatoric analysis on the majority of the multi-source pool without extrapolation.

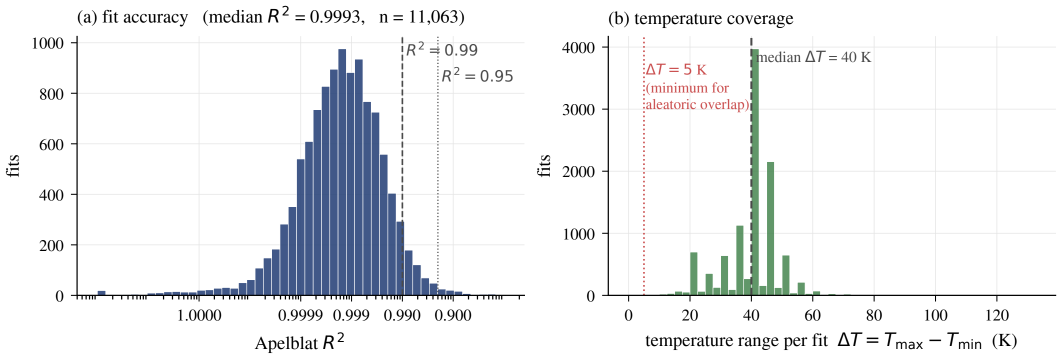

Figure 29: Apelblat fit quality. **(a)** R^2 distribution on a reversed log scale ($1 - R^2$ axis). Most fits have $R^2 > 0.99$. **(b)** Temperature coverage ΔT per fit. Most fits cover more than the 5 K minimum needed for aleatoric interpolation; the median fit spans 30 K.